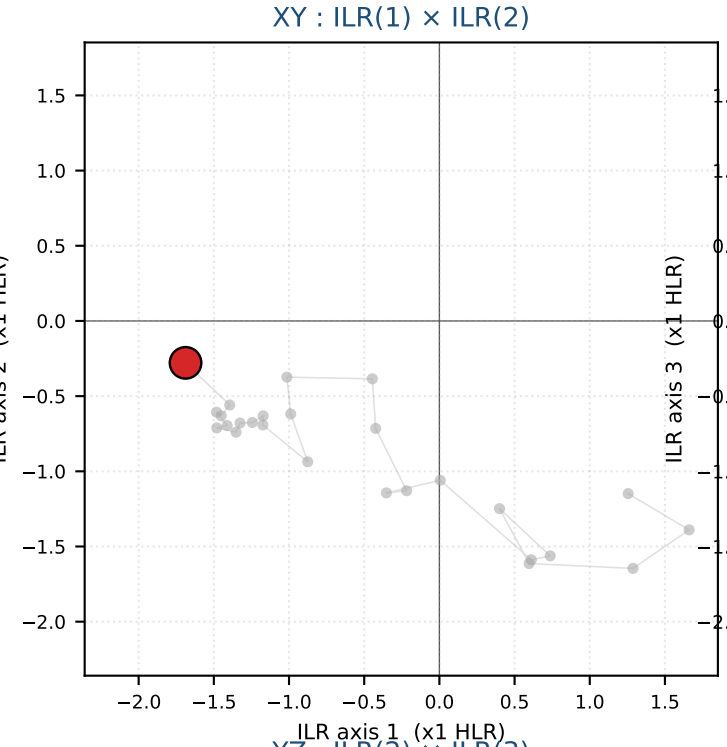
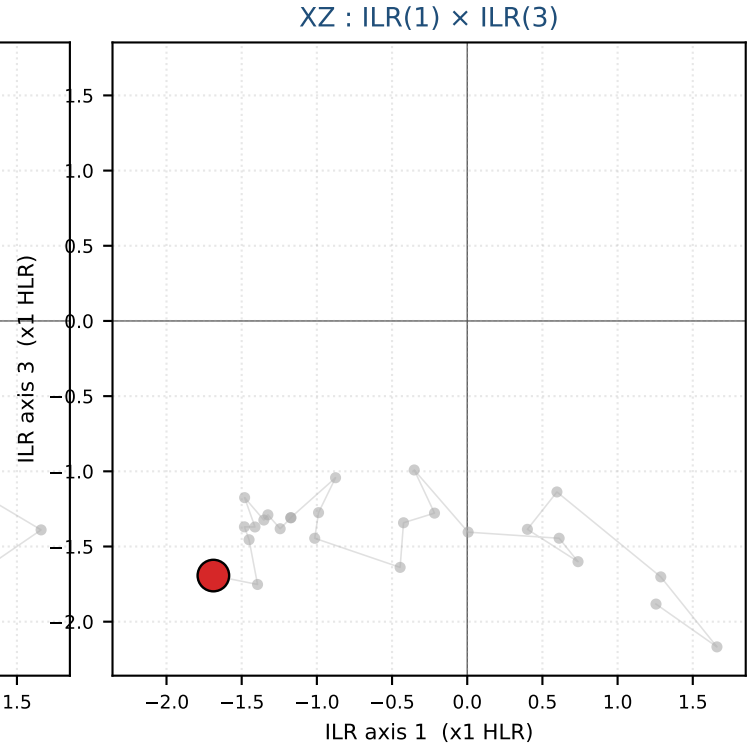
STAGE 1 v4 | t=0 | 2000 | D=9 T=26 | level 1 first-principles

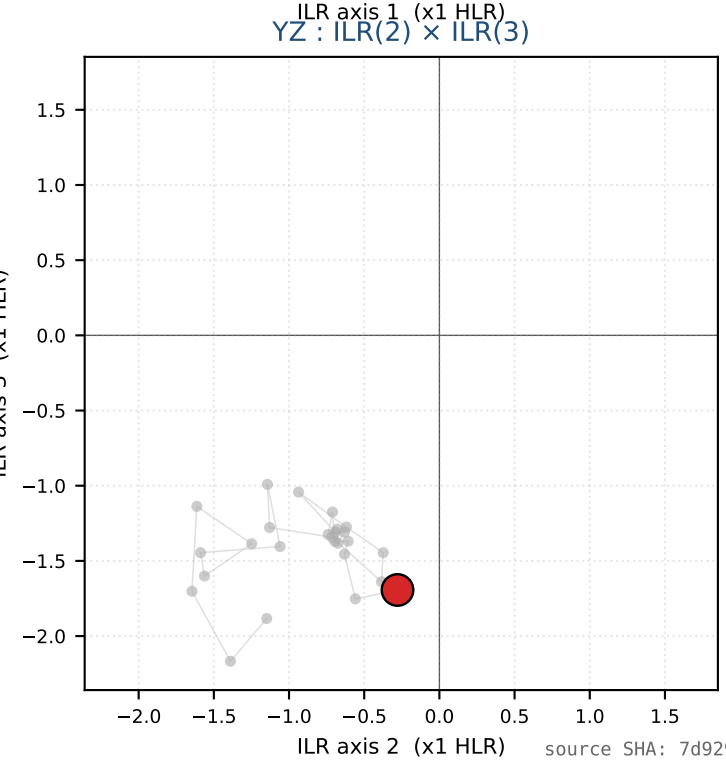
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



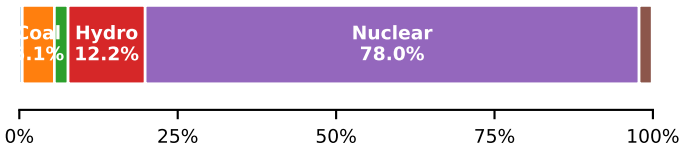


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0047	-0.311
Coal	0.0507	+2.076
Gas	0.0216	+1.224
Hydro	0.1216	+2.951
Nuclear	0.7796	+4.809
Other Fossil	0.0208	+1.184
Other Renewabl	0.0010	-1.893
Solar	0.0000	-5.825
Wind	0.0001	-4.215



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

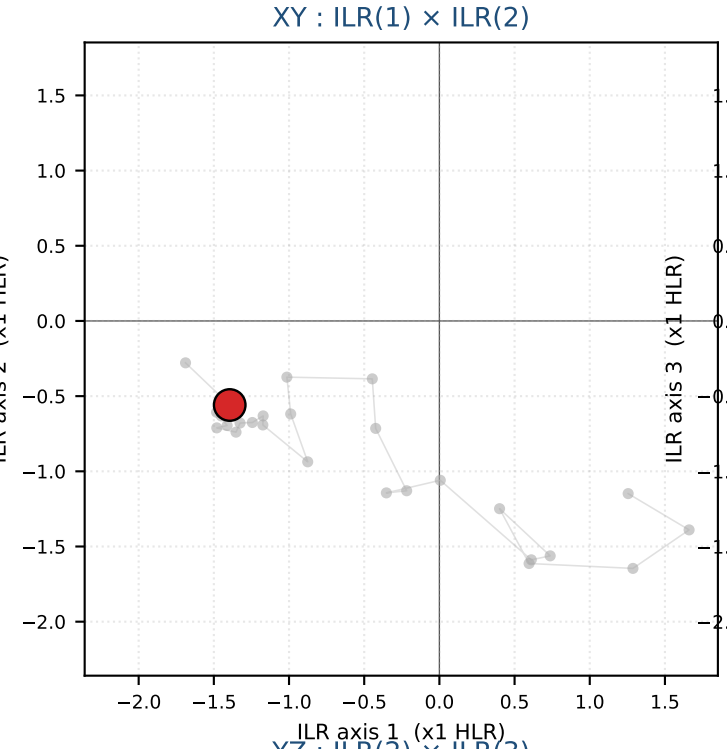
display factor: x1 (raw HLR x 1)

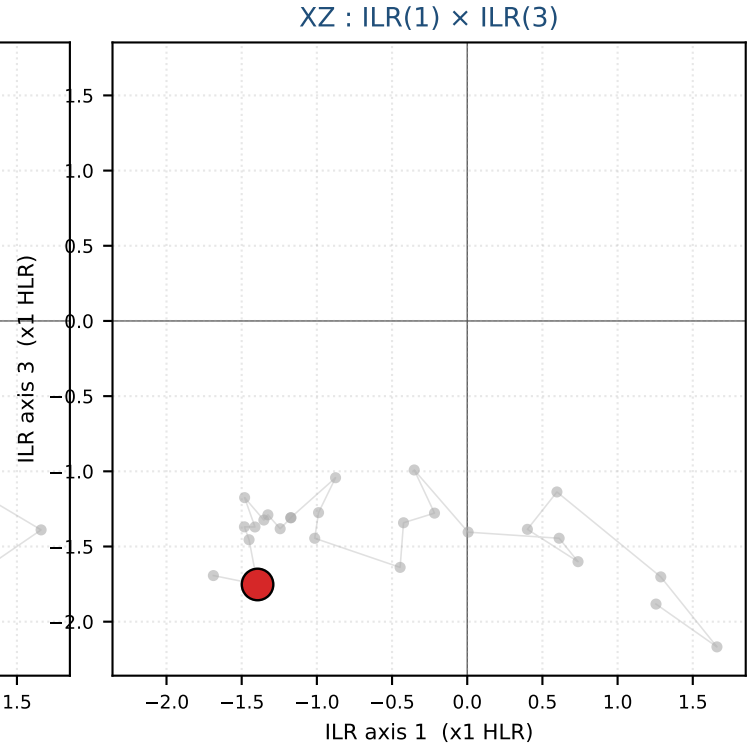
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=1 | 2001 | D=9 T=26 | level 1 first-principles

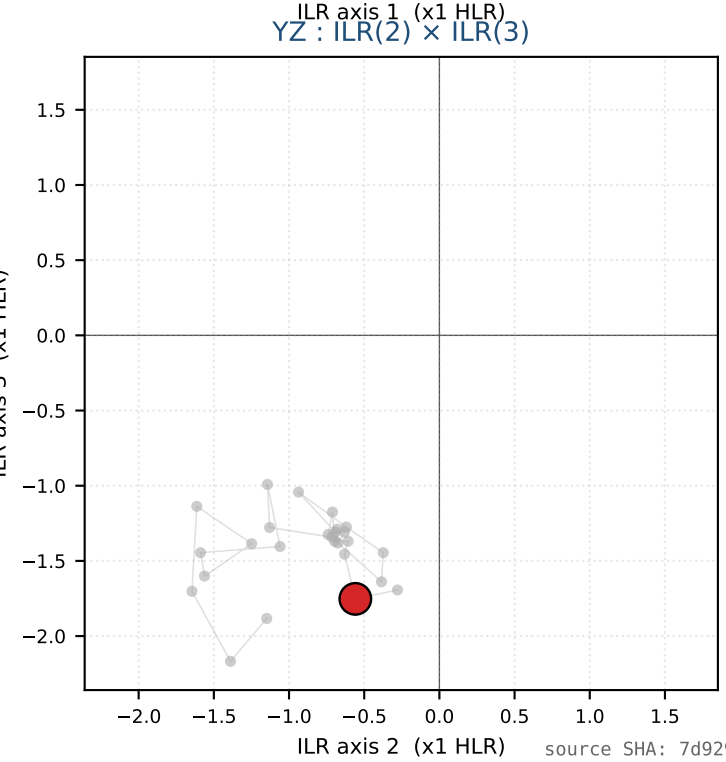
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

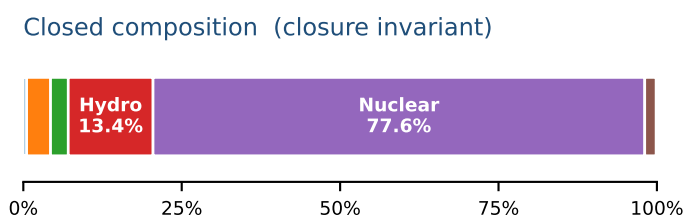




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0053	-0.288
Coal	0.0377	+1.684
Gas	0.0279	+1.383
Hydro	0.1338	+2.950
Nuclear	0.7760	+4.708
Other Fossil	0.0181	+0.947
Other Renewabl	0.0009	-2.069
Solar	0.0000	-5.940
Wind	0.0002	-3.375





closure: Σ = 1.000000 OK

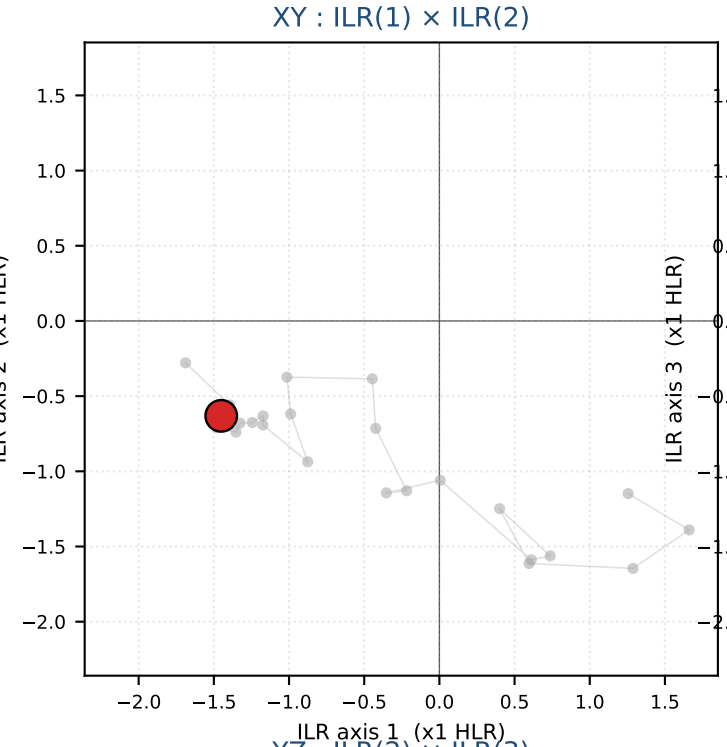
display factor: x1 (raw HLR x 1)

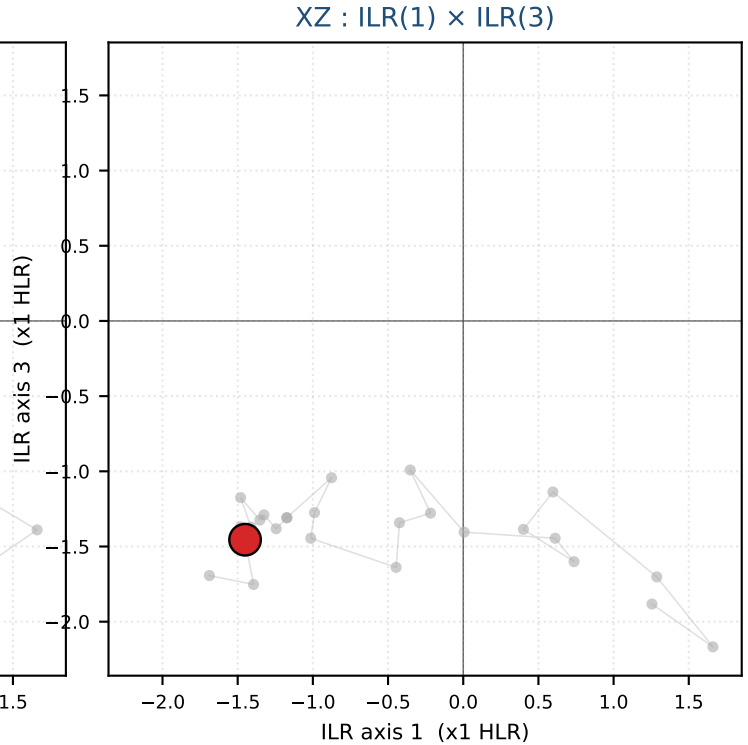
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=2 | 2002 | D=9 T=26 | level 1 first-principles

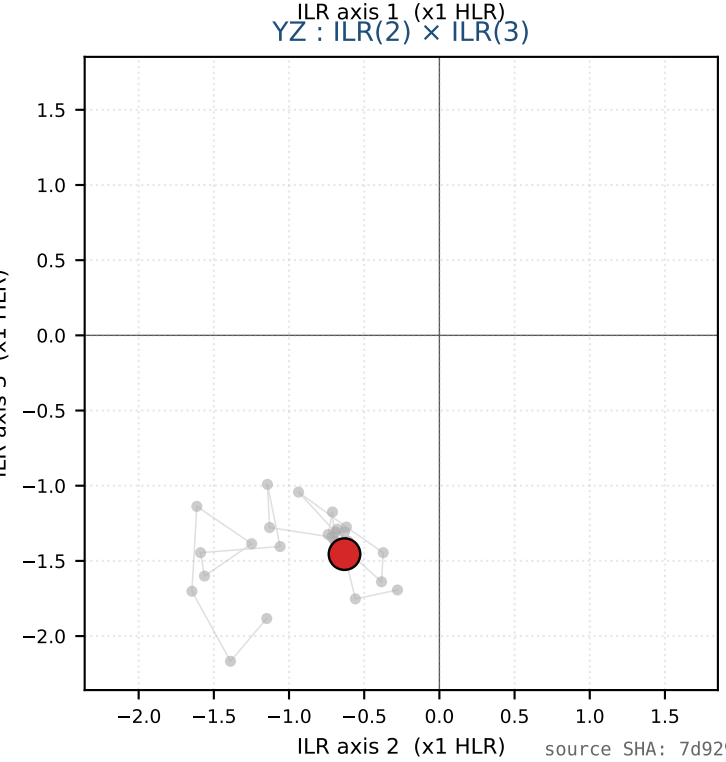
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

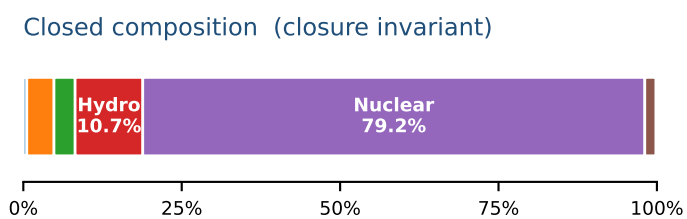




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0055	-0.332
Coal	0.0429	+1.720
Gas	0.0333	+1.467
Hydro	0.1068	+2.632
Nuclear	0.7923	+4.636
Other Fossil	0.0177	+0.835
Other Renewabl	0.0009	-2.157
Solar	0.0000	-6.049
Wind	0.0005	-2.753





closure: Σ = 1.000000 OK

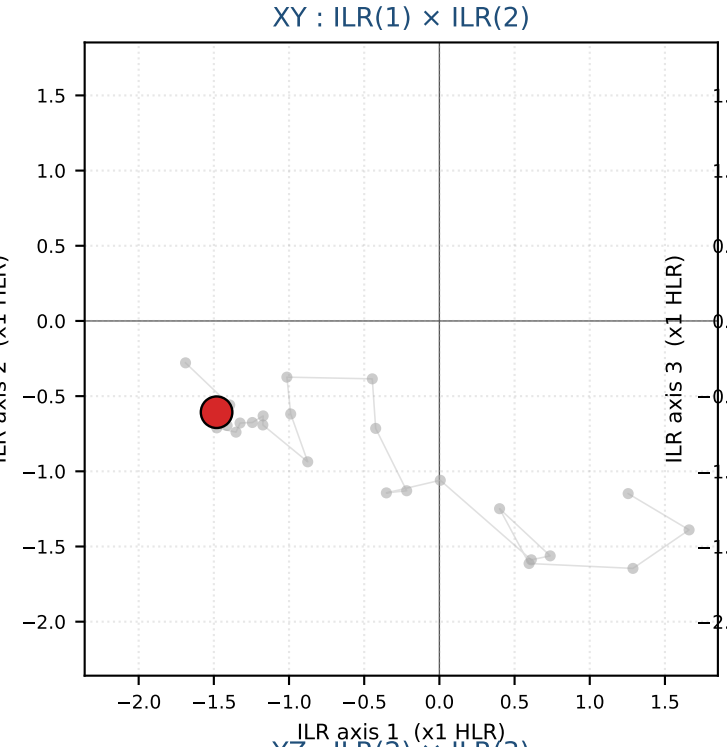
display factor: x1 (raw HLR x 1)

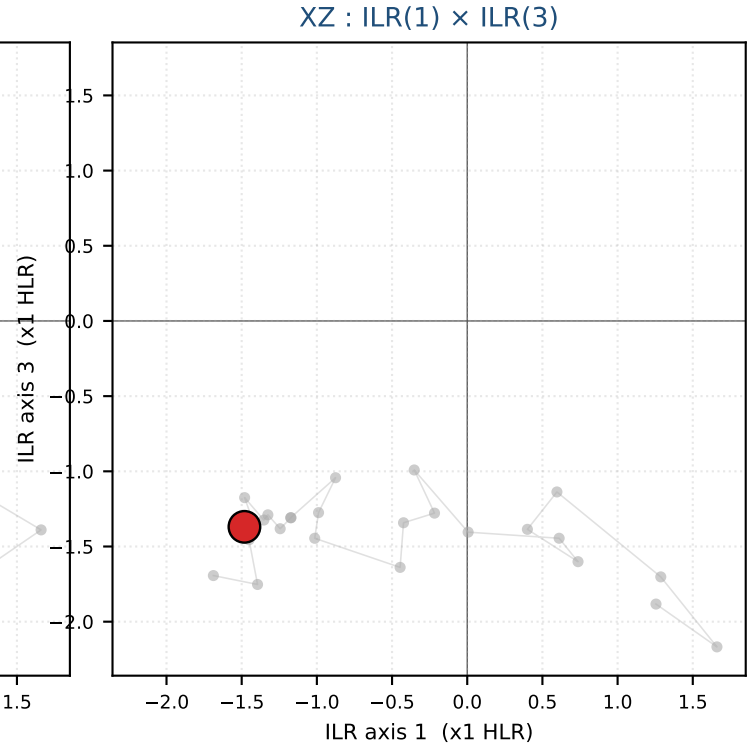
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=3 | 2003 | D=9 T=26 | level 1 first-principles

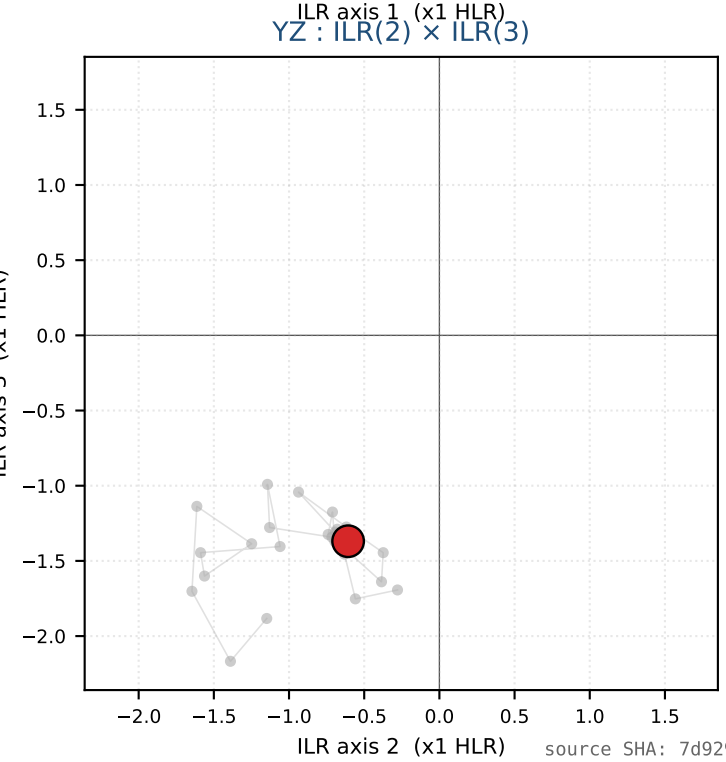
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



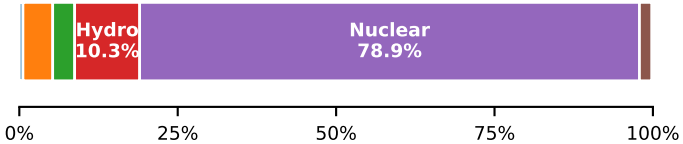


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0058	-0.347
Coal	0.0470	+1.748
Gas	0.0346	+1.444
Hydro	0.1026	+2.530
Nuclear	0.7889	+4.570
Other Fossil	0.0196	+0.875
Other Renewabl	0.0009	-2.233
Solar	0.0000	-6.125
Wind	0.0007	-2.461



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

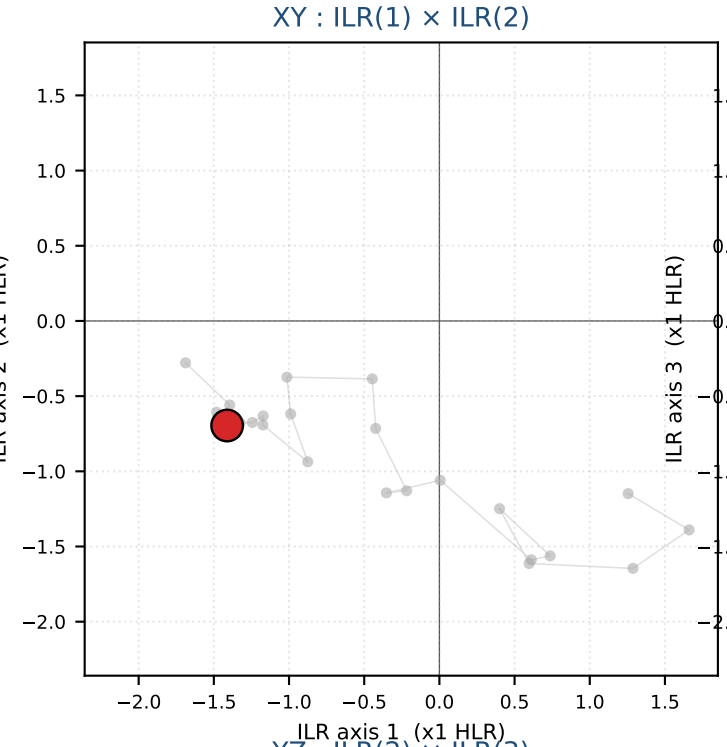
display factor: x1 (raw HLR x 1)

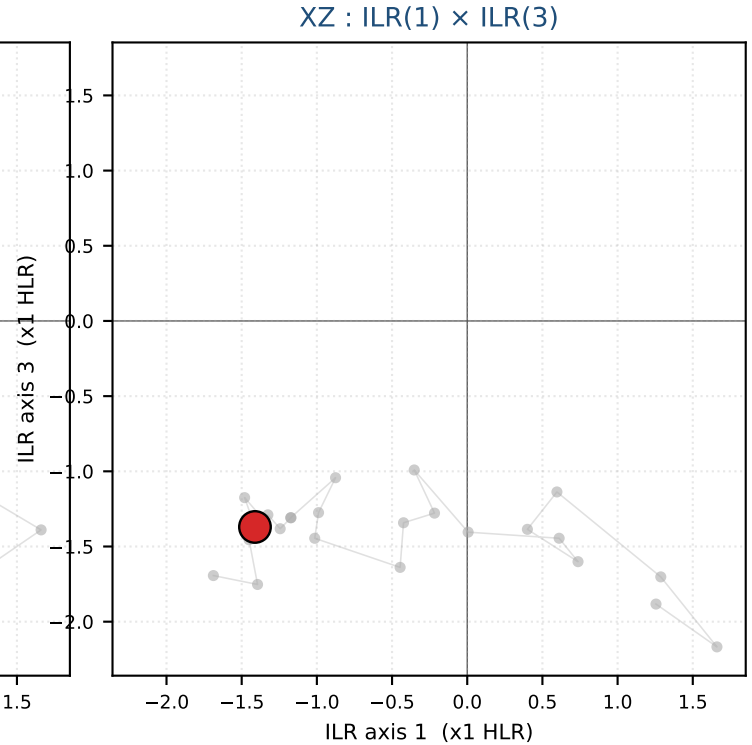
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=4 | 2004 | D=9 T=26 | level 1 first-principles

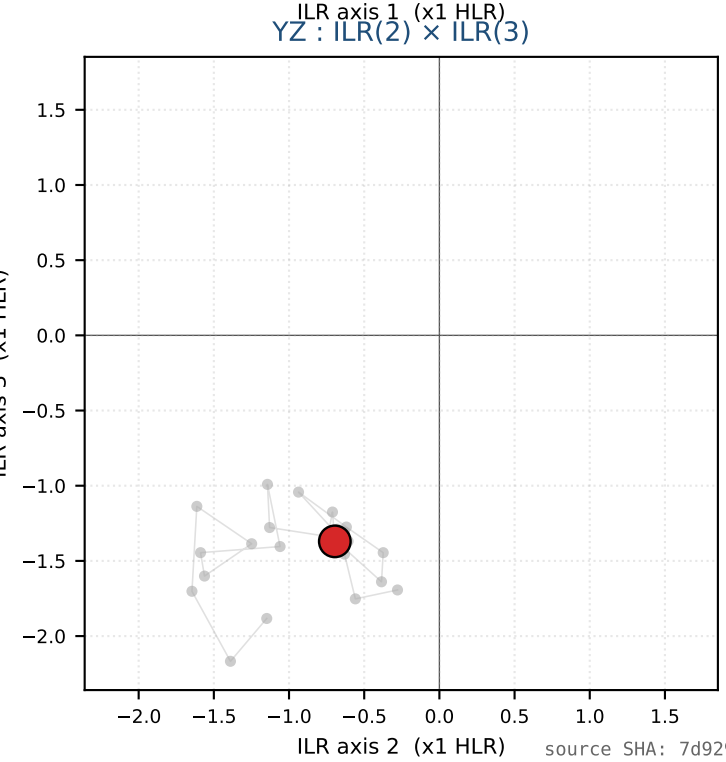
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

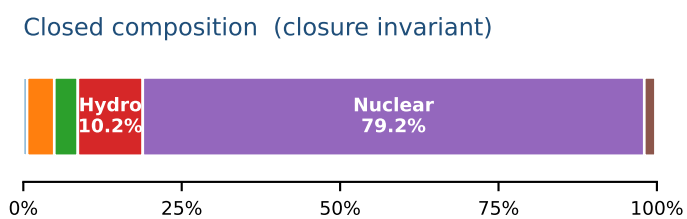




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0058	-0.363
Coal	0.0430	+1.633
Gas	0.0372	+1.487
Hydro	0.1025	+2.501
Nuclear	0.7915	+4.545
Other Fossil	0.0180	+0.764
Other Renewabl	0.0008	-2.315
Solar	0.0000	-6.165
Wind	0.0010	-2.088





closure: Σ = 1.000000 OK

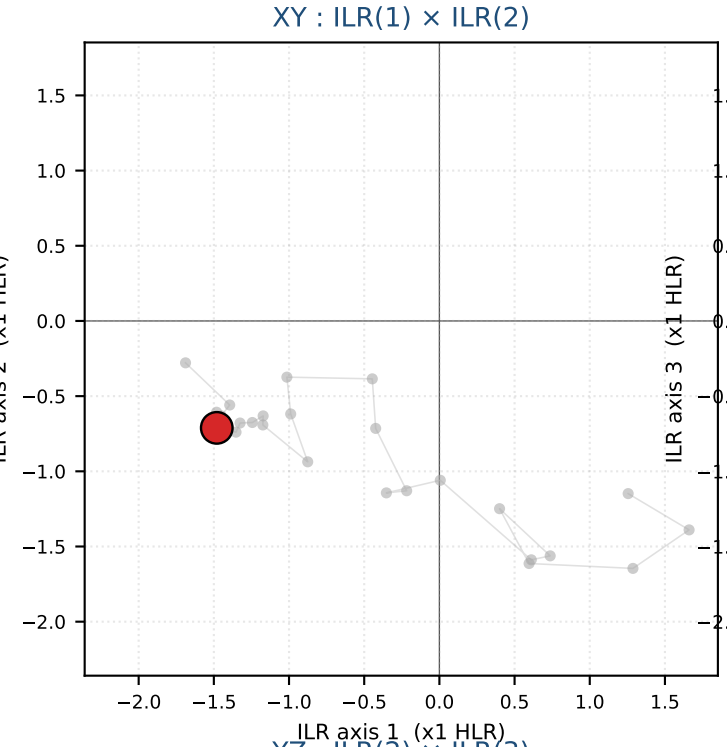
display factor: x1 (raw HLR x 1)

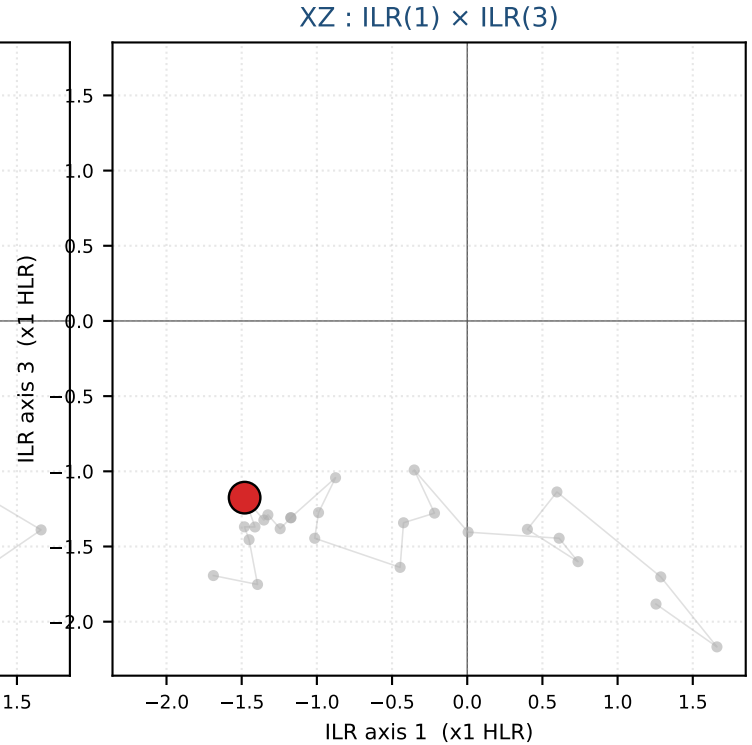
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=5 | 2005 | D=9 T=26 | level 1 first-principles

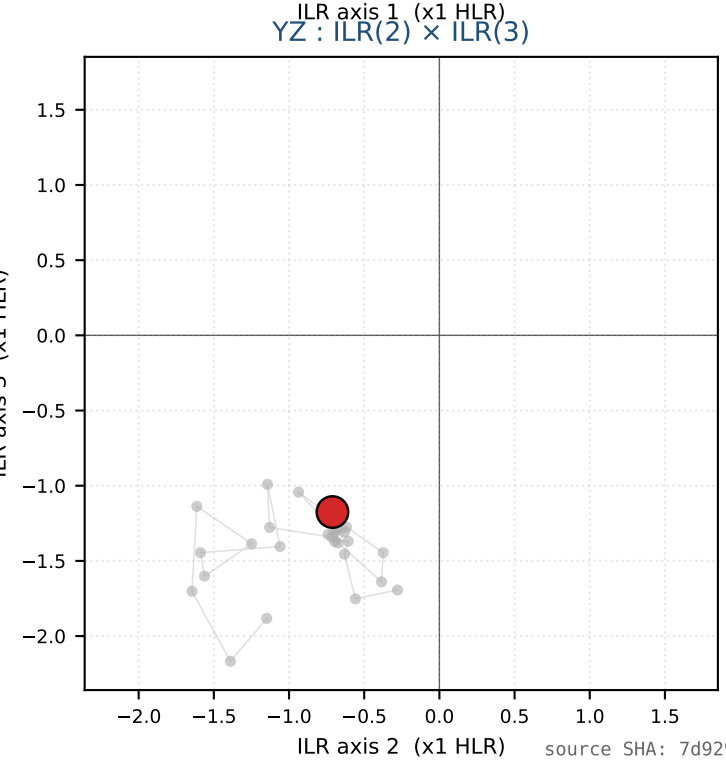
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

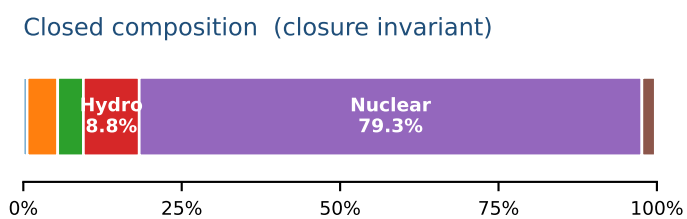




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0060	-0.425
Coal	0.0483	+1.669
Gas	0.0405	+1.493
Hydro	0.0881	+2.269
Nuclear	0.7934	+4.467
Other Fossil	0.0212	+0.843
Other Renewabl	0.0008	-2.380
Solar	0.0000	-6.251
Wind	0.0017	-1.686





closure: Σ = 1.000000 OK

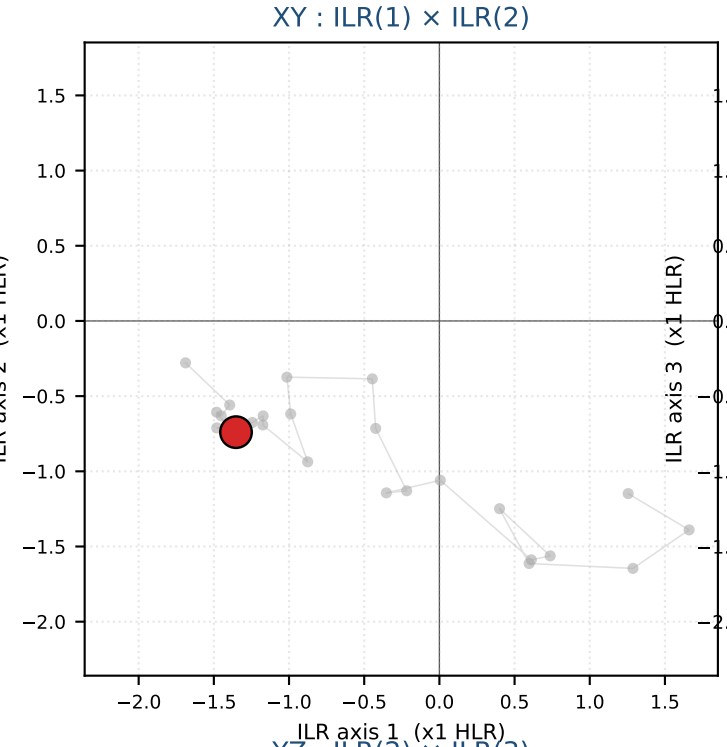
display factor: x1 (raw HLR x 1)

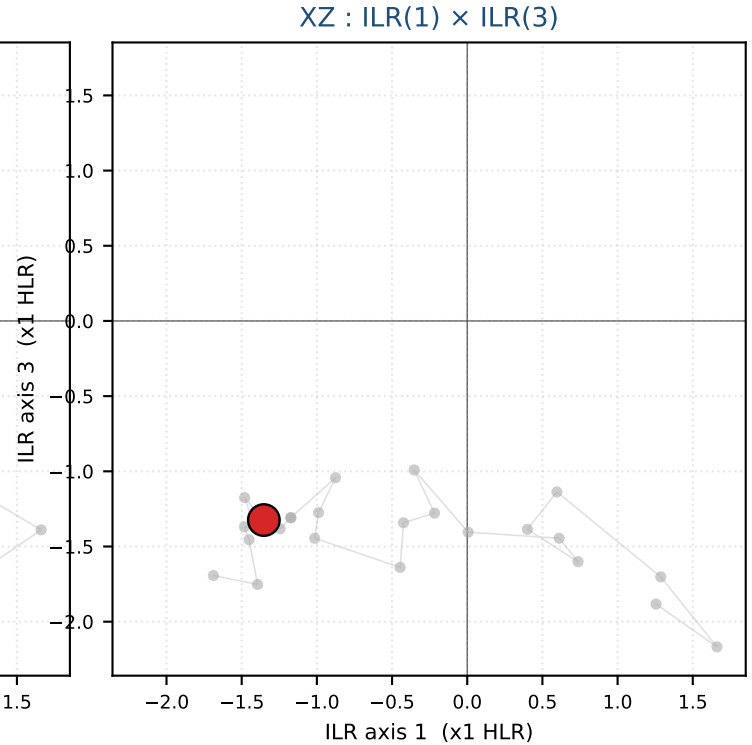
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=6 | 2006 | D=9 T=26 | level 1 first-principles

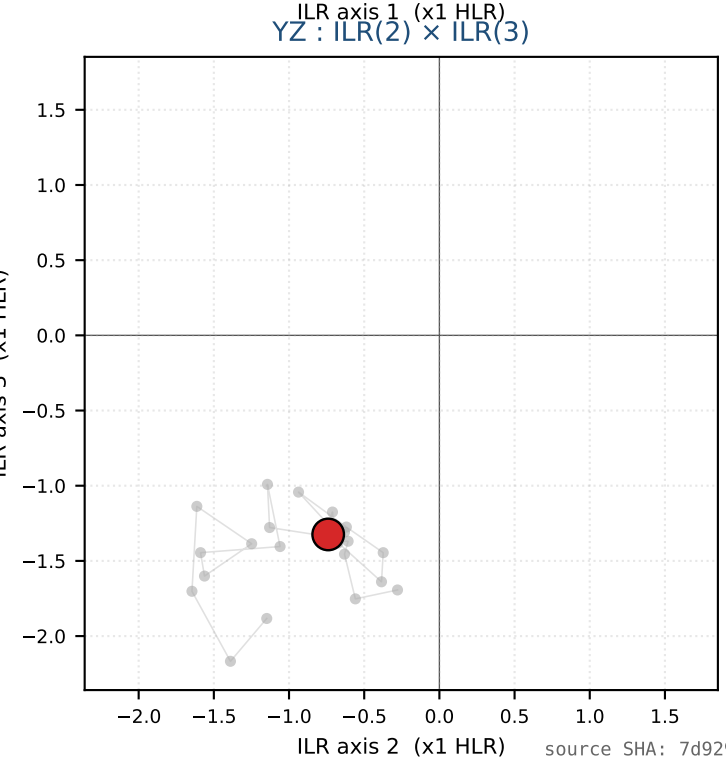
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

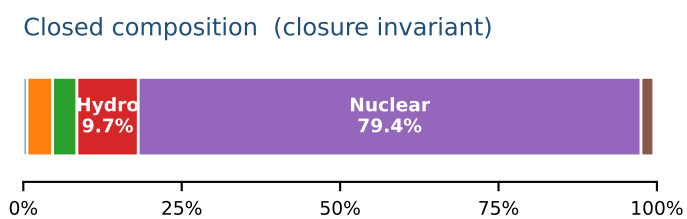




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0060	-0.492
Coal	0.0403	+1.421
Gas	0.0384	+1.370
Hydro	0.0967	+2.295
Nuclear	0.7936	+4.400
Other Fossil	0.0204	+0.737
Other Renewabl	0.0008	-2.486
Solar	0.0000	-6.315
Wind	0.0038	-0.930





closure: Σ = 1.000000 OK

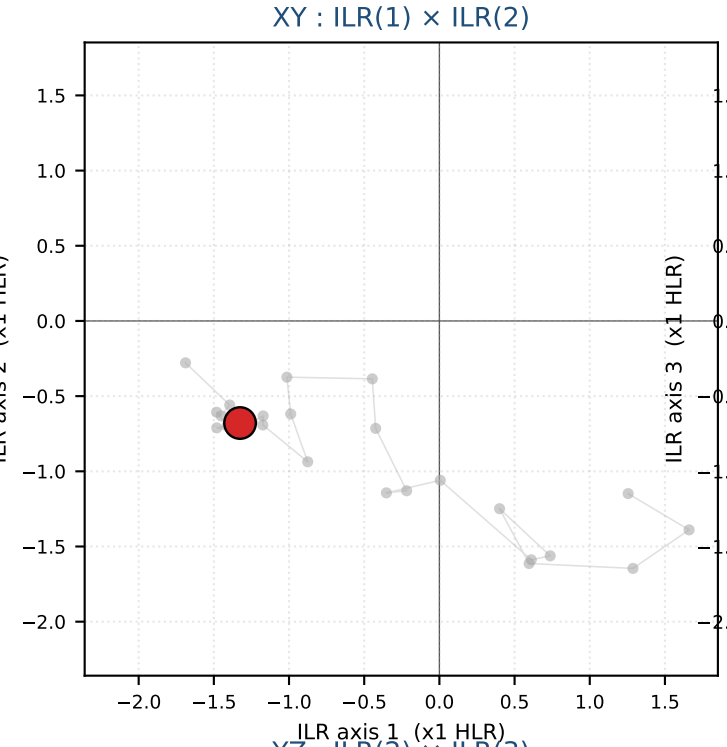
display factor: x1 (raw HLR x 1)

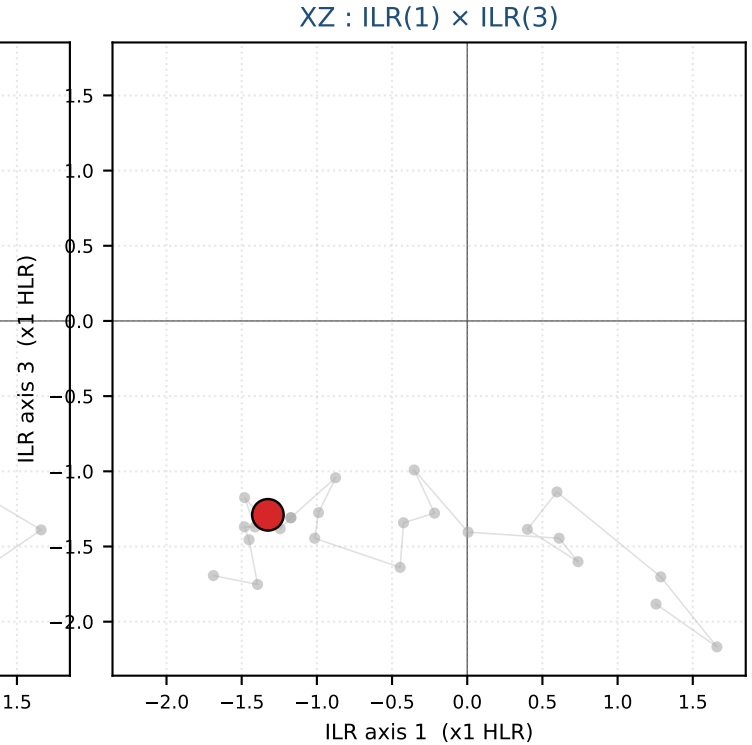
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=7 | 2007 | D=9 T=26 | level 1 first-principles

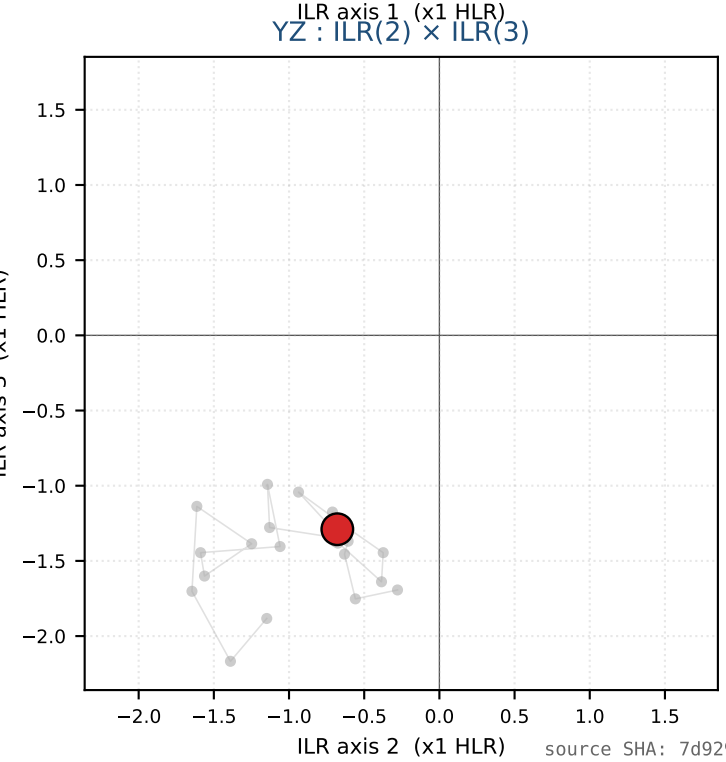
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

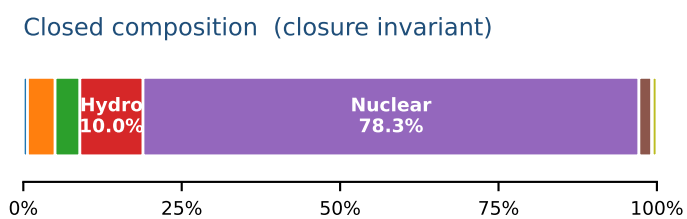




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0067	-0.555
Coal	0.0435	+1.320
Gas	0.0391	+1.214
Hydro	0.0997	+2.149
Nuclear	0.7827	+4.210
Other Fossil	0.0202	+0.555
Other Renewabl	0.0008	-2.631
Solar	0.0000	-5.788
Wind	0.0072	-0.473





closure: Σ = 1.000000 OK

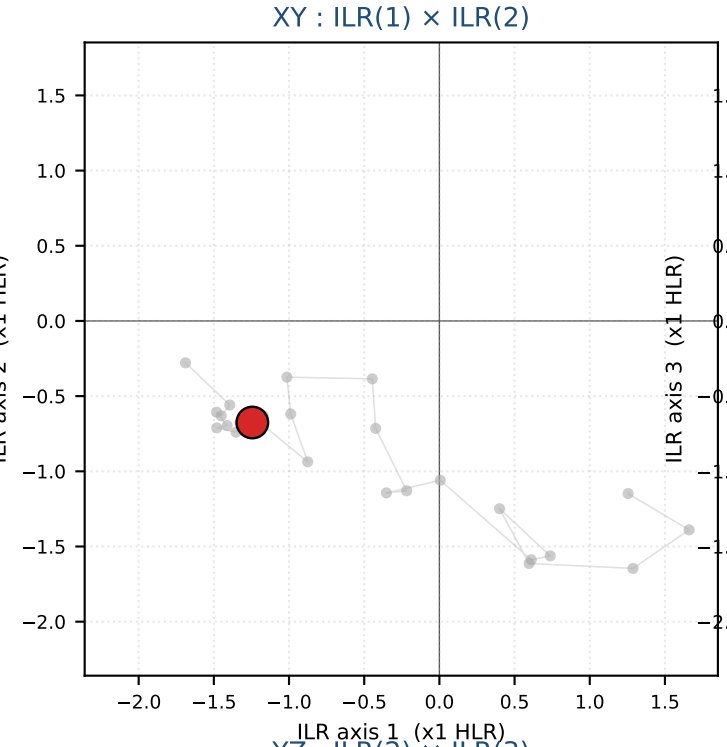
display factor: x1 (raw HLR x 1)

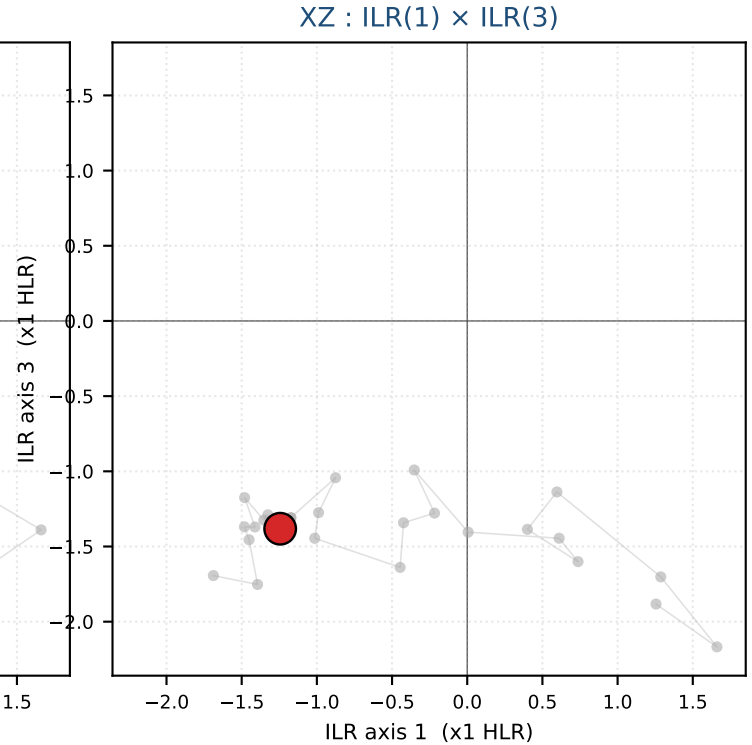
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=8 | 2008 | D=9 T=26 | level 1 first-principles

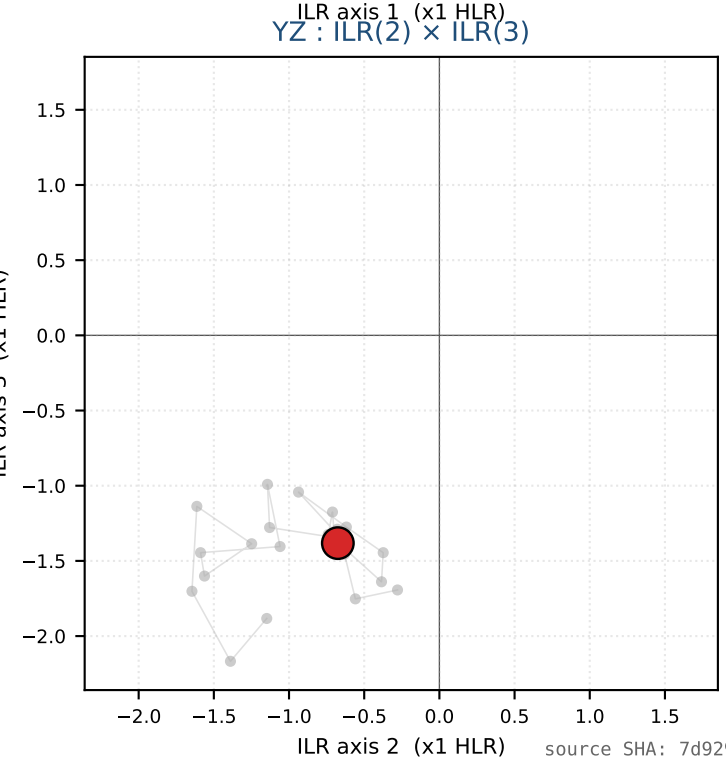
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



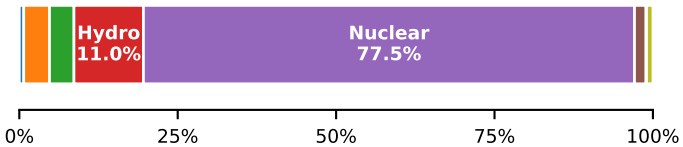


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0070	-0.611
Coal	0.0407	+1.148
Gas	0.0386	+1.095
Hydro	0.1096	+2.140
Nuclear	0.7751	+4.095
Other Fossil	0.0181	+0.338
Other Renewabl	0.0008	-2.745
Solar	0.0001	-5.209
Wind	0.0100	-0.251



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

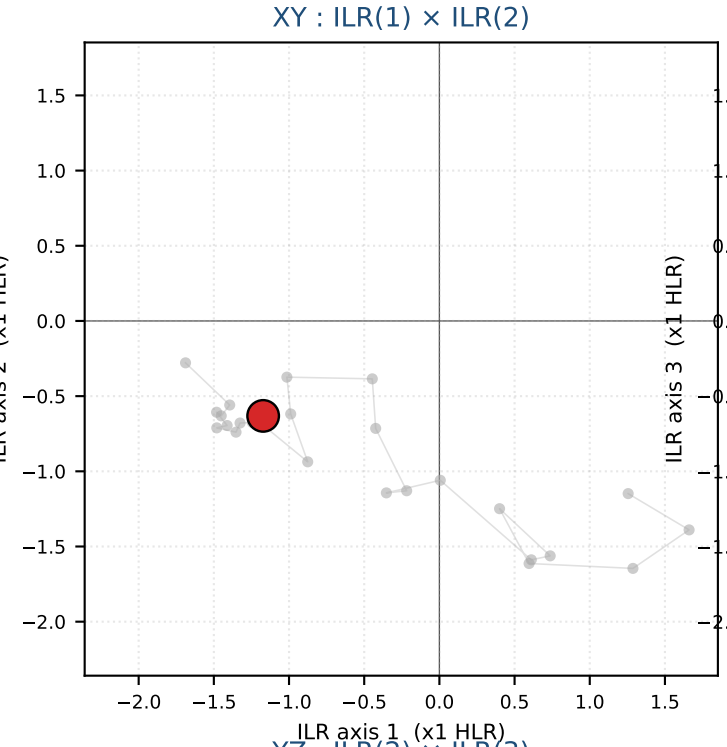
display factor: x1 (raw HLR x 1)

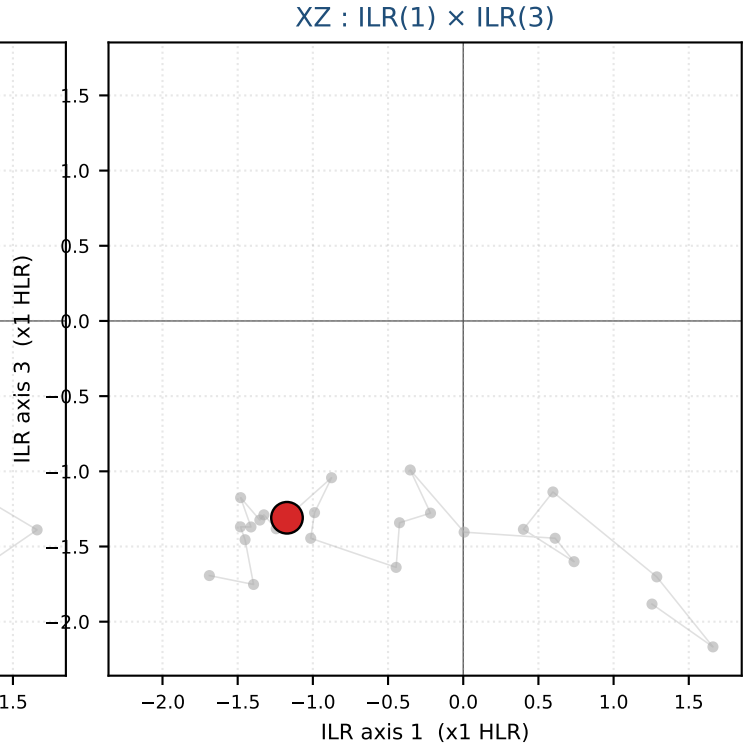
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=9 | 2009 | D=9 T=26 | level 1 first-principles

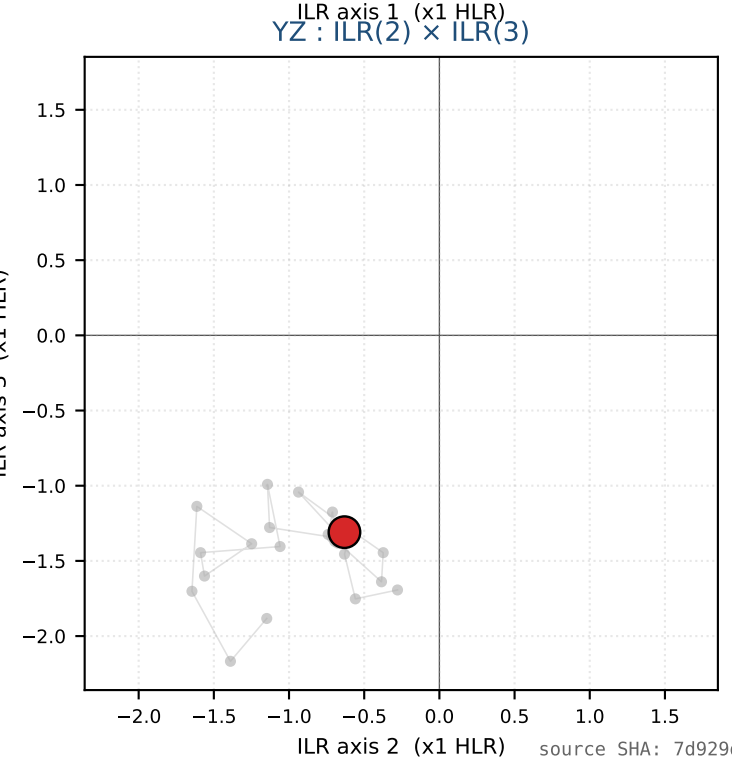
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

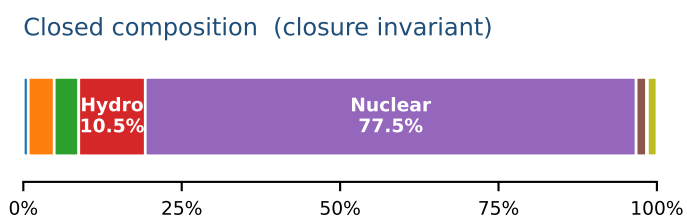




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0078	-0.717
Coal	0.0410	+0.940
Gas	0.0388	+0.885
Hydro	0.1049	+1.881
Nuclear	0.7747	+3.880
Other Fossil	0.0167	+0.040
Other Renewabl	0.0009	-2.934
Solar	0.0003	-3.908
Wind	0.0150	-0.067





closure: Σ = 1.000000 OK

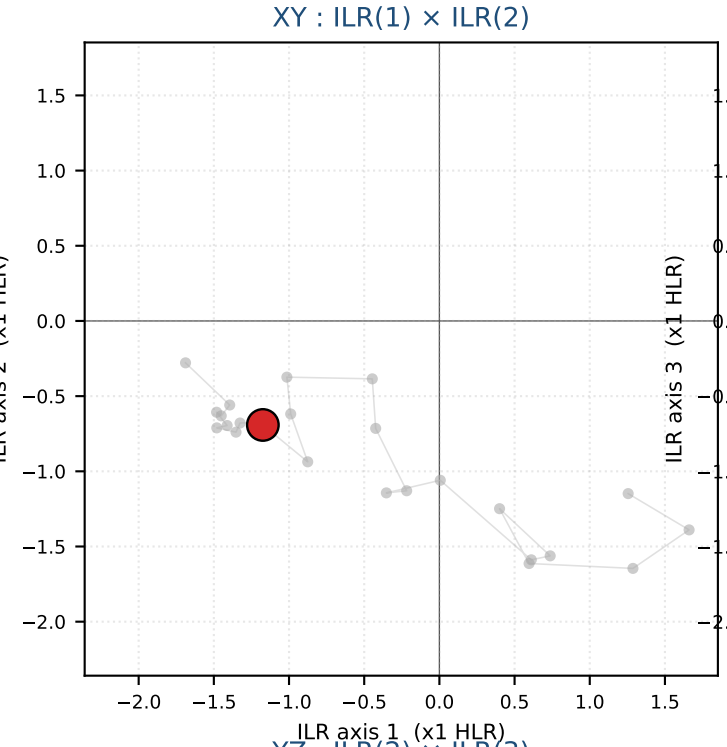
display factor: x1 (raw HLR x 1)

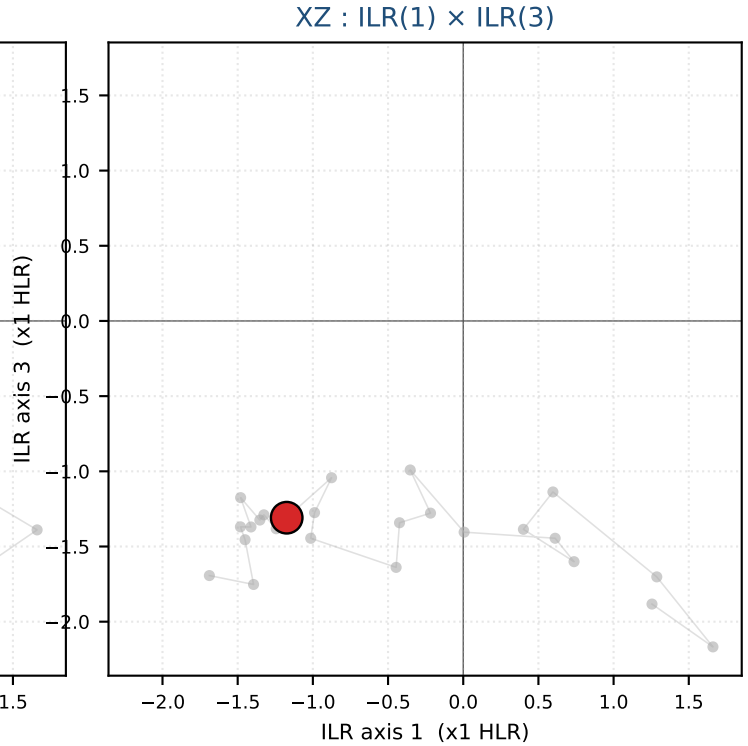
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=10 | 2010 | D=9 T=26 | level 1 first-principles

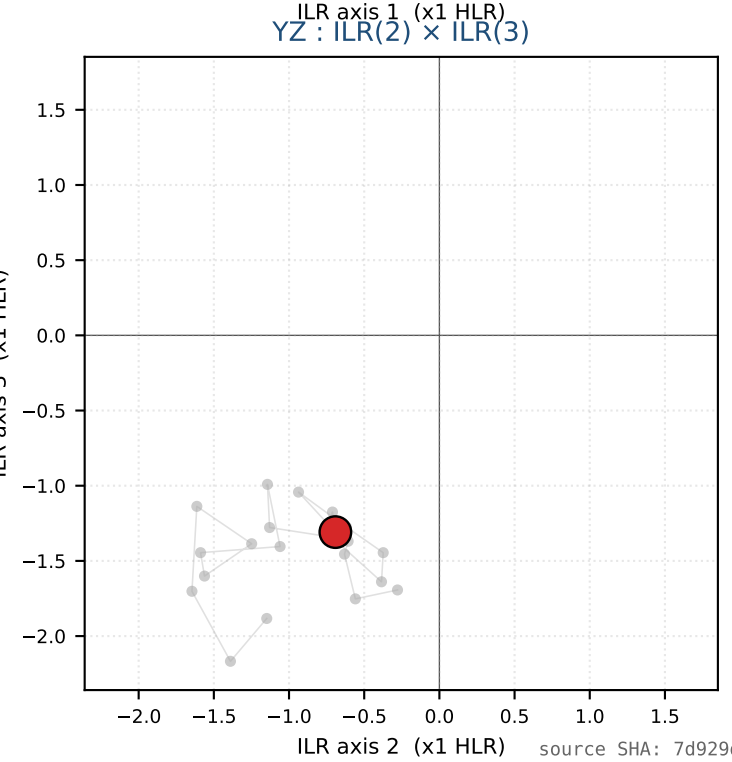
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

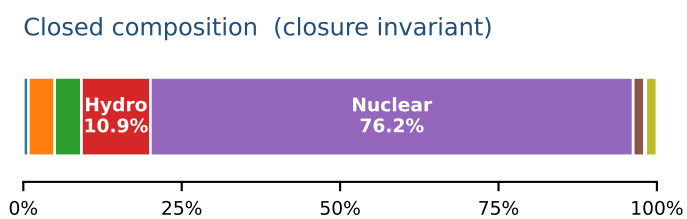




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0079	-0.884
Coal	0.0415	+0.776
Gas	0.0423	+0.793
Hydro	0.1088	+1.739
Nuclear	0.7620	+3.685
Other Fossil	0.0179	-0.068
Other Renewabl	0.0009	-3.109
Solar	0.0011	-2.853
Wind	0.0177	-0.078





closure: Σ = 1.000000 OK

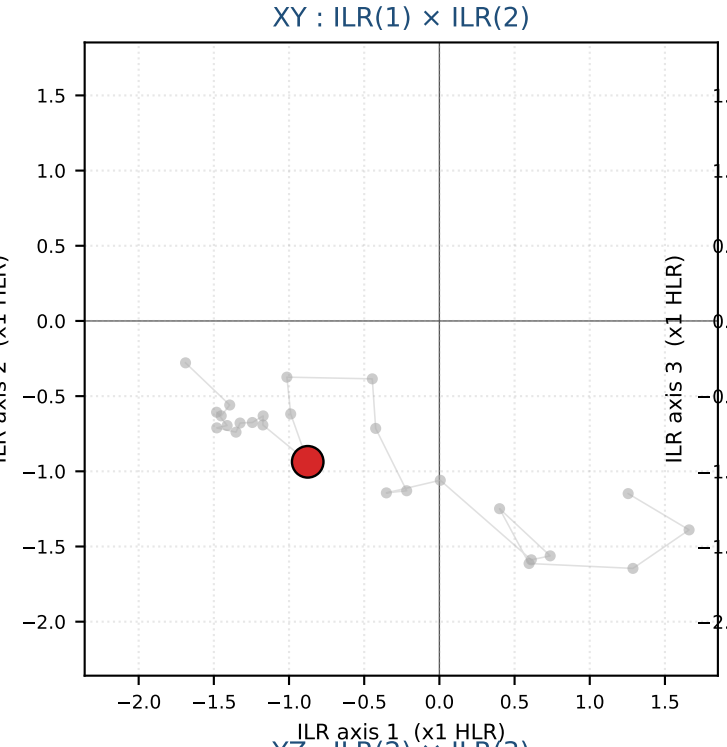
display factor: x1 (raw HLR x 1)

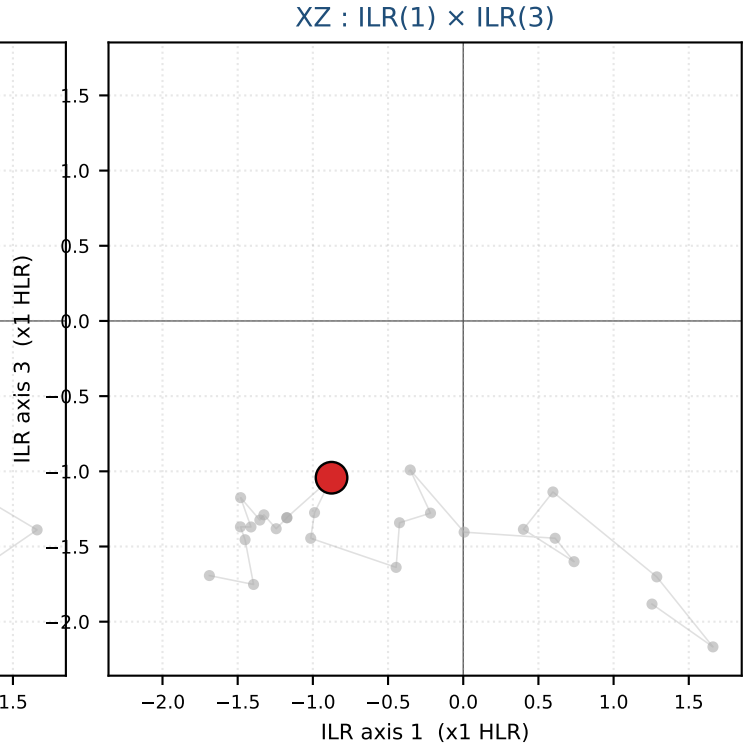
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=11 | 2011 | D=9 T=26 | level 1 first-principles

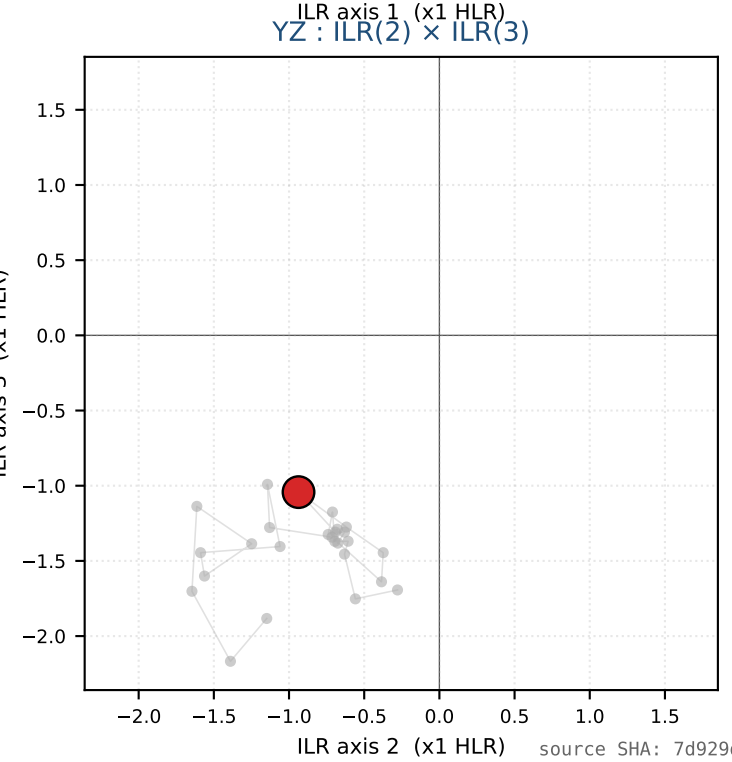
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

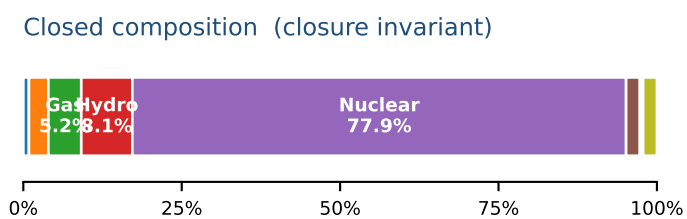




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0089	-0.942
Coal	0.0306	+0.297
Gas	0.0520	+0.825
Hydro	0.0806	+1.264
Nuclear	0.7793	+3.533
Other Fossil	0.0218	-0.045
Other Renewabl	0.0010	-3.175
Solar	0.0041	-1.713
Wind	0.0218	-0.044





closure: Σ = 1.000000 OK

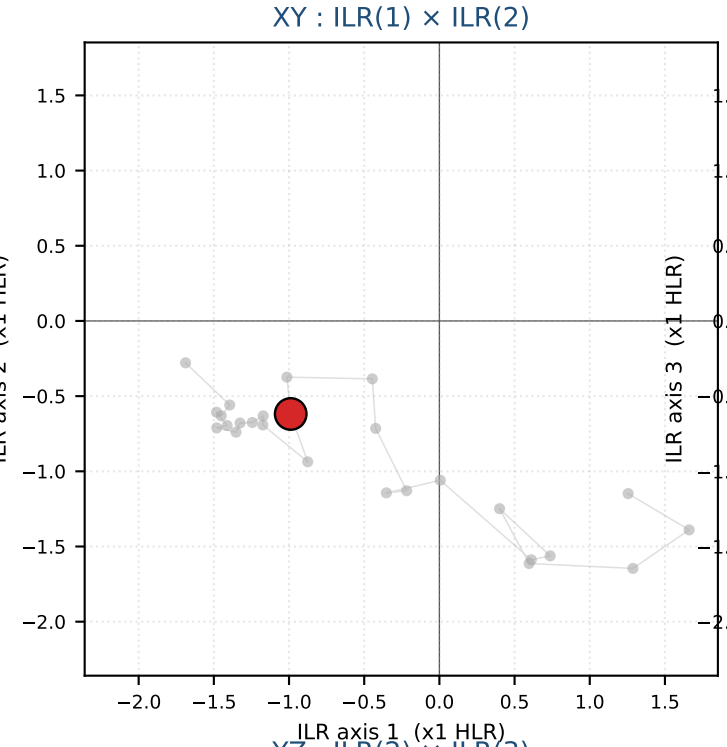
display factor: x1 (raw HLR x 1)

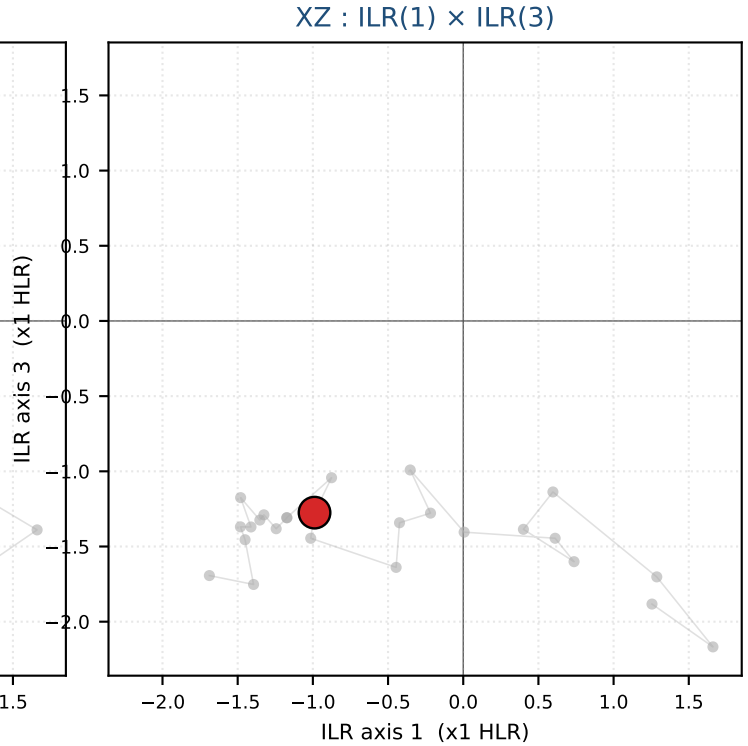
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=12 | 2012 | D=9 T=26 | level 1 first-principles

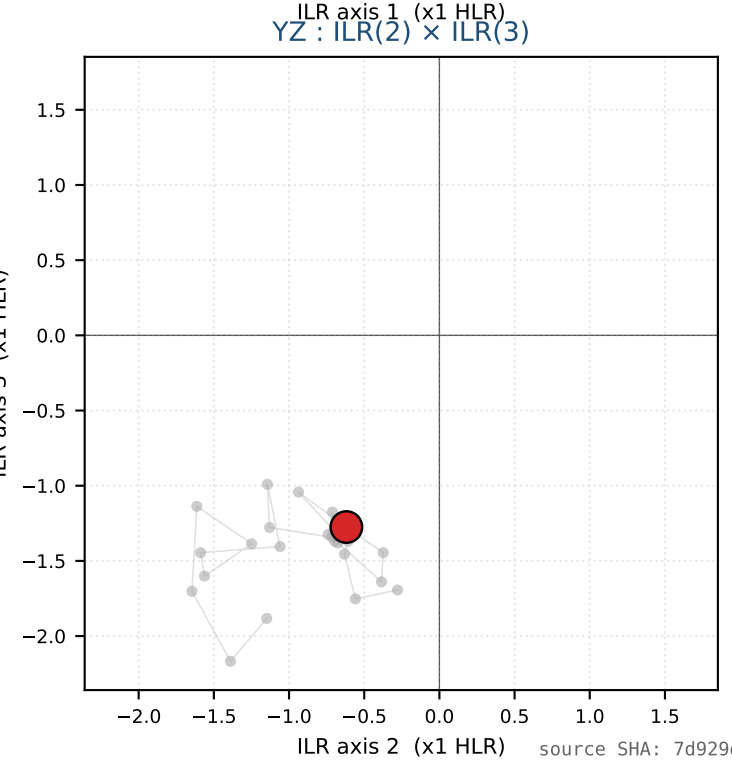
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

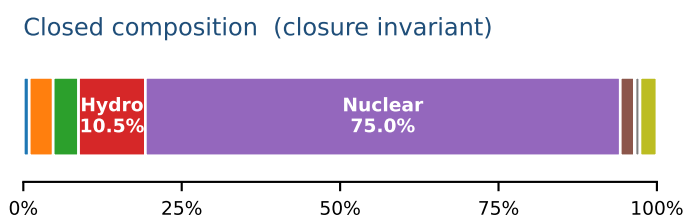




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0093	-1.008
Coal	0.0378	+0.391
Gas	0.0401	+0.449
Hydro	0.1054	+1.416
Nuclear	0.7495	+3.377
Other Fossil	0.0224	-0.133
Other Renewabl	0.0009	-3.349
Solar	0.0078	-1.187
Wind	0.0267	+0.044





closure: Σ = 1.000000 OK

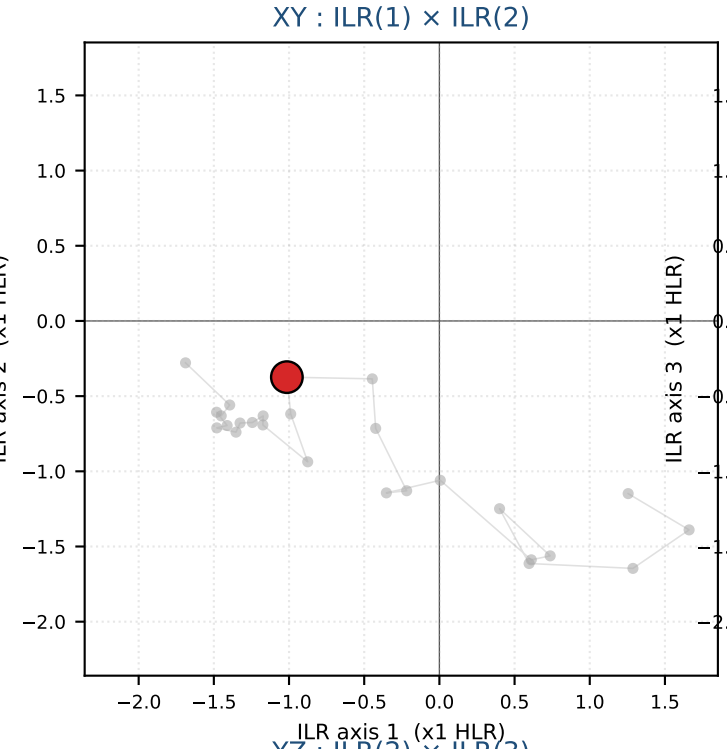
display factor: x1 (raw HLR x 1)

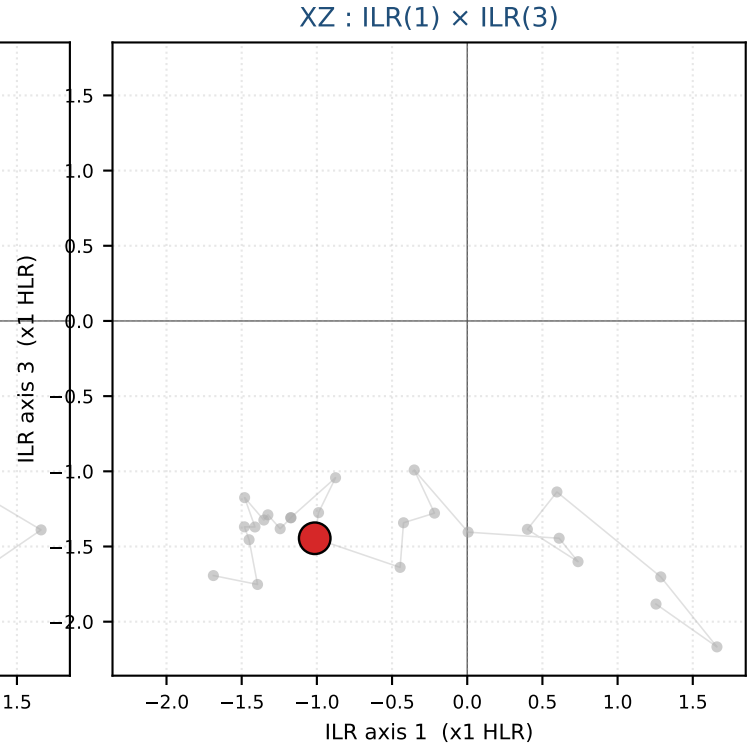
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=13 | 2013 | D=9 T=26 | level 1 first-principles

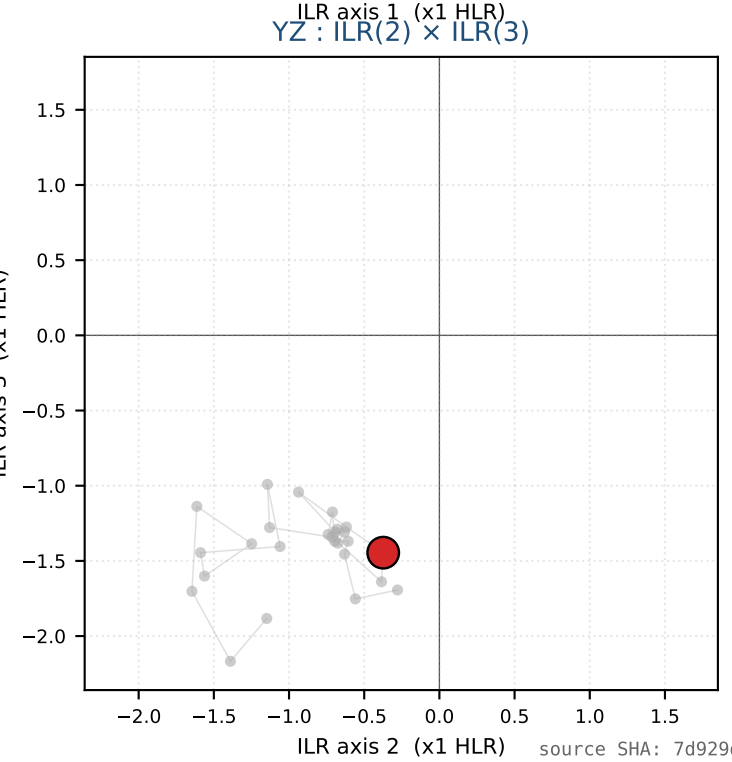
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



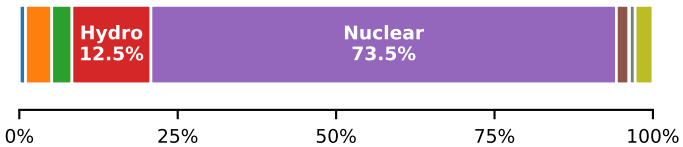


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0098	-0.965
Coal	0.0413	+0.470
Gas	0.0319	+0.210
Hydro	0.1247	+1.574
Nuclear	0.7346	+3.347
Other Fossil	0.0198	-0.267
Other Renewabl	0.0009	-3.395
Solar	0.0090	-1.055
Wind	0.0280	+0.080



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

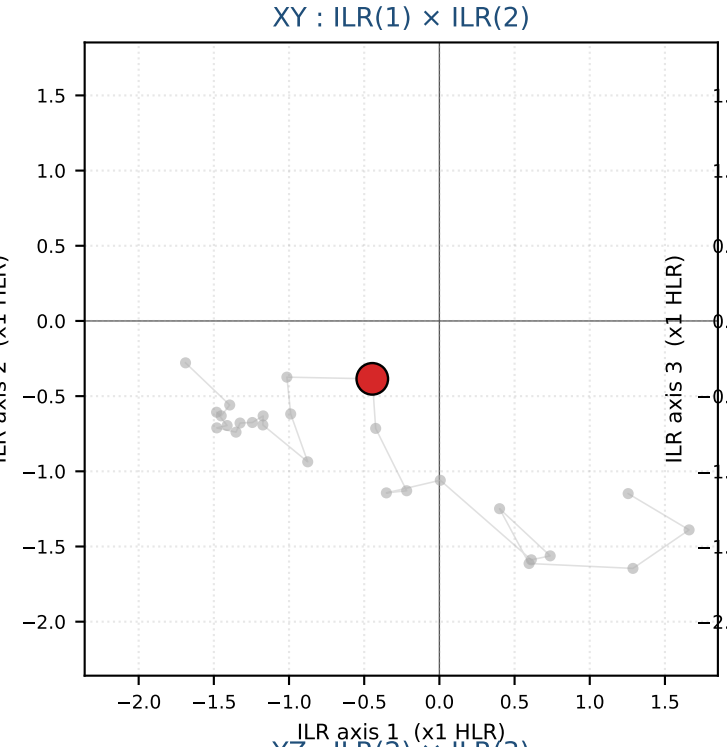
display factor: x1 (raw HLR x 1)

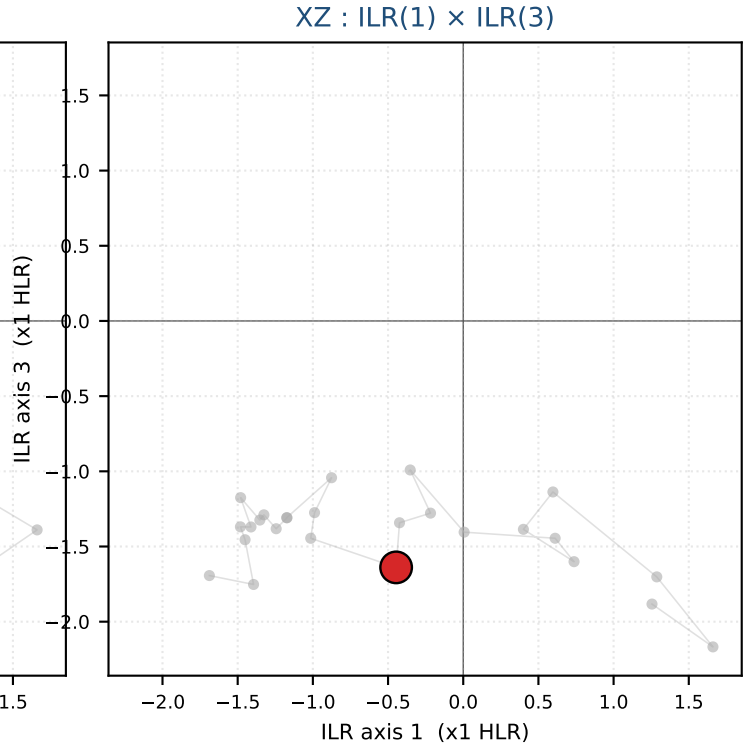
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=14 | 2014 | D=9 T=26 | level 1 first-principles

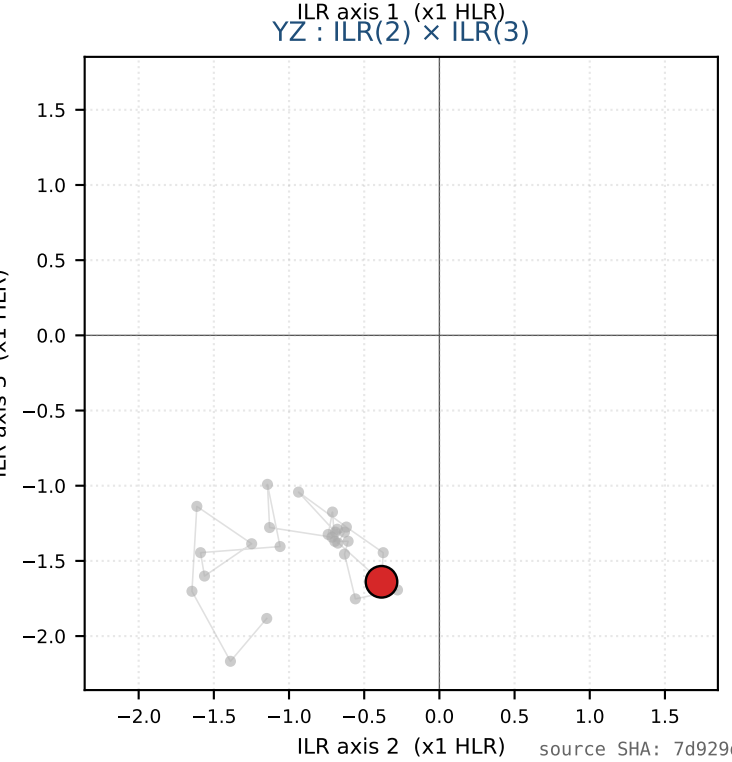
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

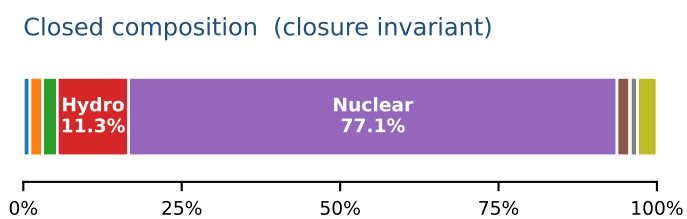




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0106	-0.830
Coal	0.0199	-0.199
Gas	0.0232	-0.043
Hydro	0.1126	+1.535
Nuclear	0.7707	+3.459
Other Fossil	0.0201	-0.188
Other Renewabl	0.0010	-3.200
Solar	0.0113	-0.765
Wind	0.0306	+0.232





closure: Σ = 1.000000 OK

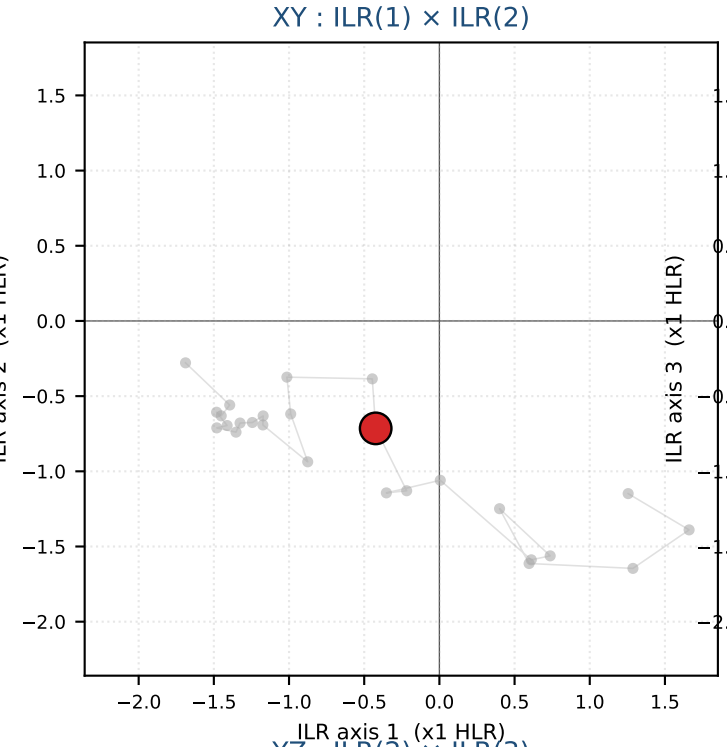
display factor: x1 (raw HLR x 1)

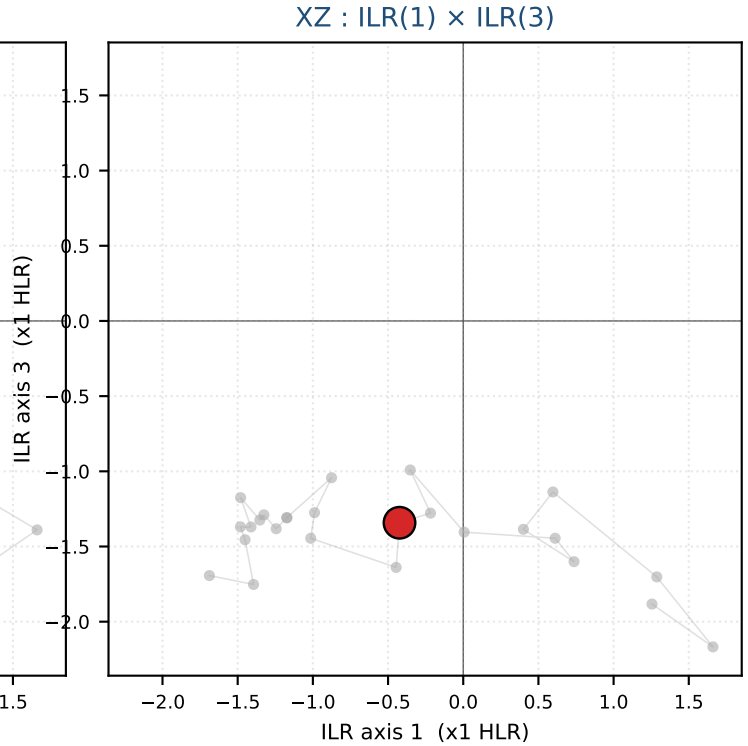
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=15 | 2015 | D=9 T=26 | level 1 first-principles

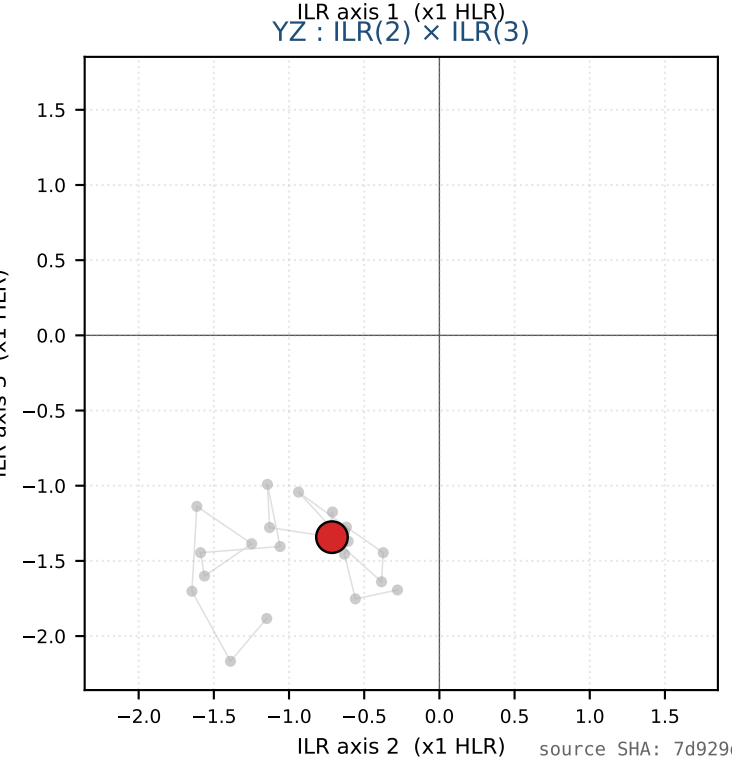
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



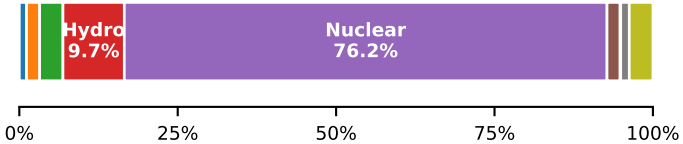


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0114	-0.850
Coal	0.0207	-0.251
Gas	0.0368	+0.324
Hydro	0.0968	+1.291
Nuclear	0.7619	+3.354
Other Fossil	0.0206	-0.255
Other Renewabl	0.0010	-3.271
Solar	0.0135	-0.679
Wind	0.0373	+0.338



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

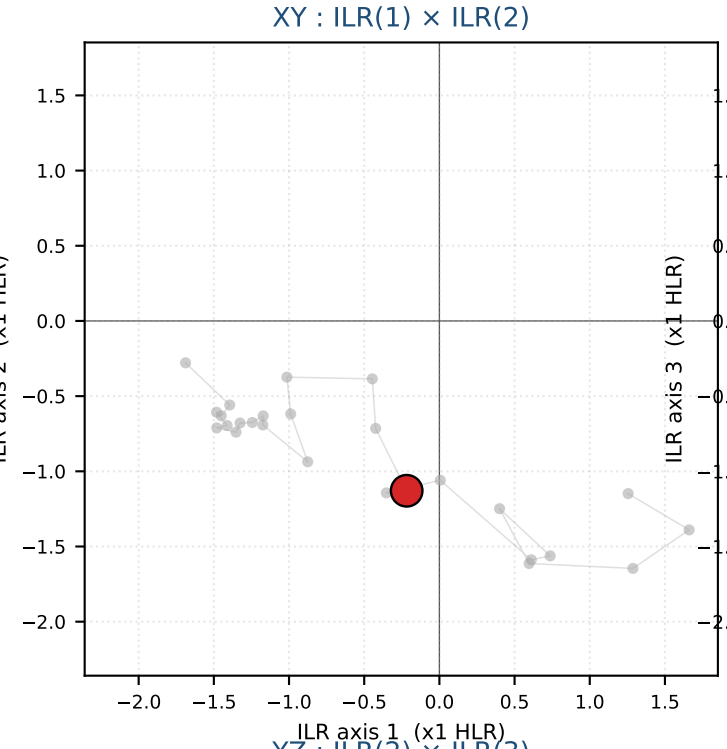
display factor: x1 (raw HLR x 1)

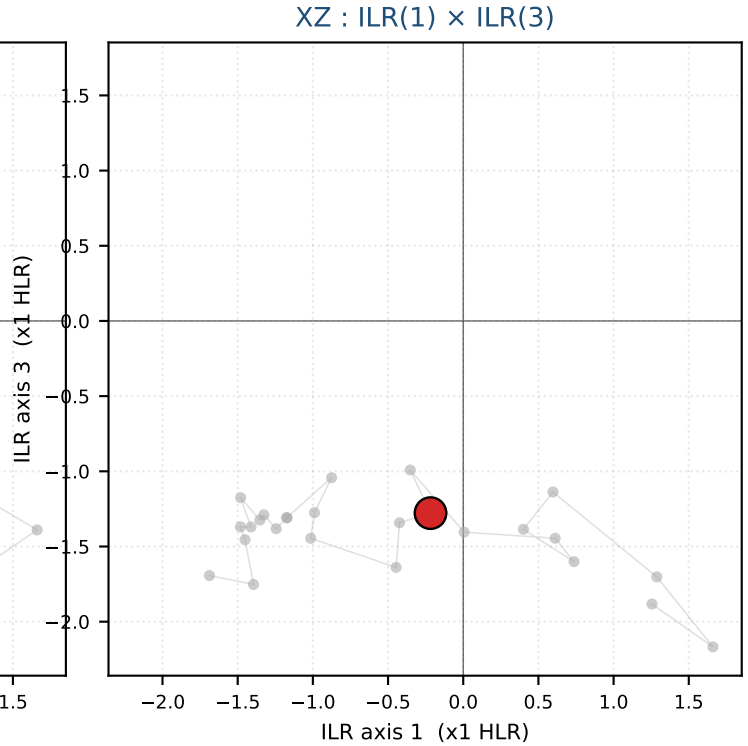
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=16 | 2016 | D=9 T=26 | level 1 first-principles

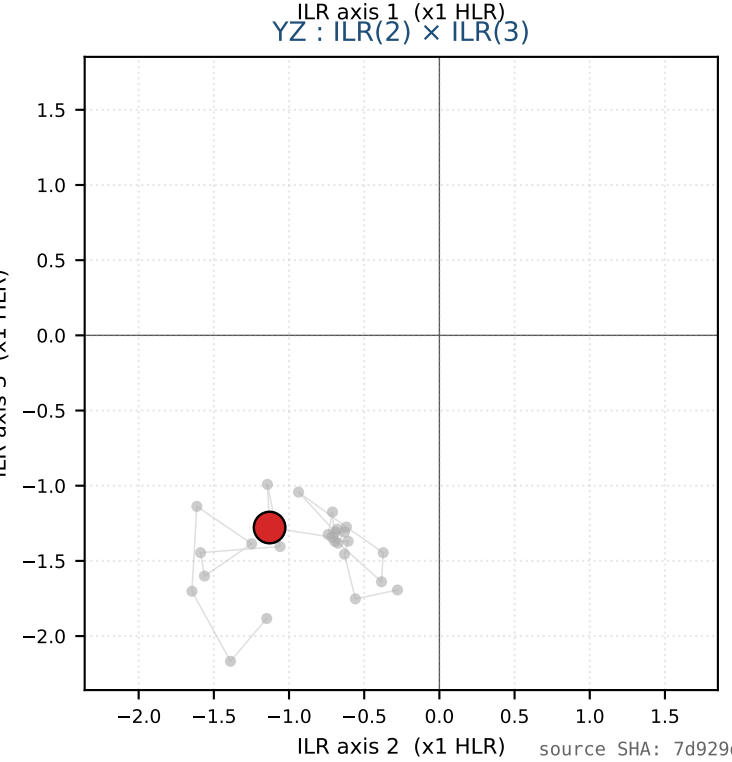
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



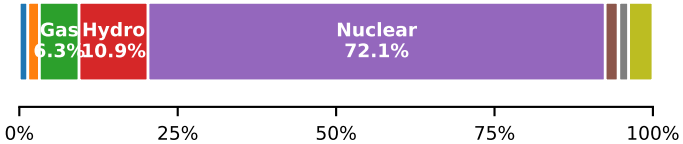


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0135	-0.777
Coal	0.0183	-0.469
Gas	0.0626	+0.760
Hydro	0.1089	+1.314
Nuclear	0.7215	+3.205
Other Fossil	0.0205	-0.358
Other Renewabl	0.0011	-3.306
Solar	0.0155	-0.636
Wind	0.0383	+0.268



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

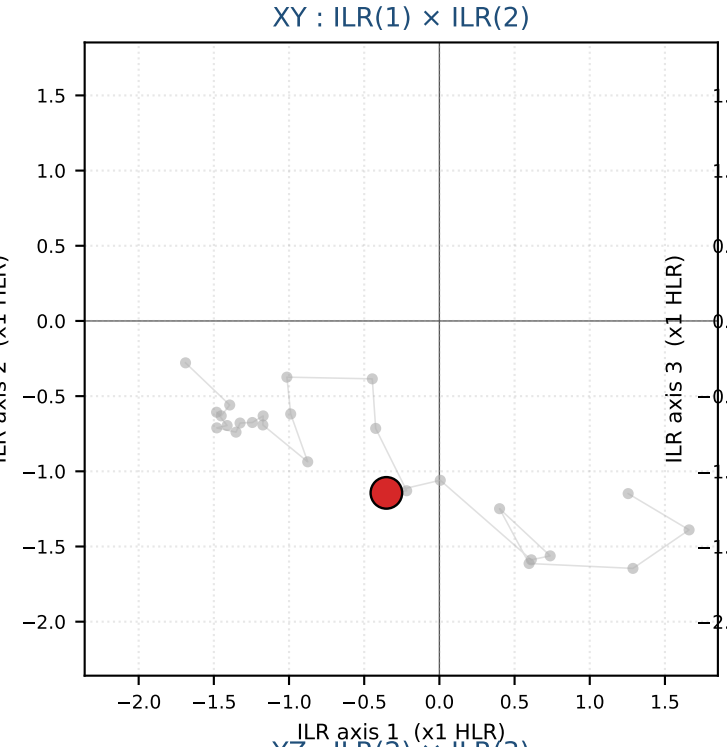
display factor: x1 (raw HLR x 1)

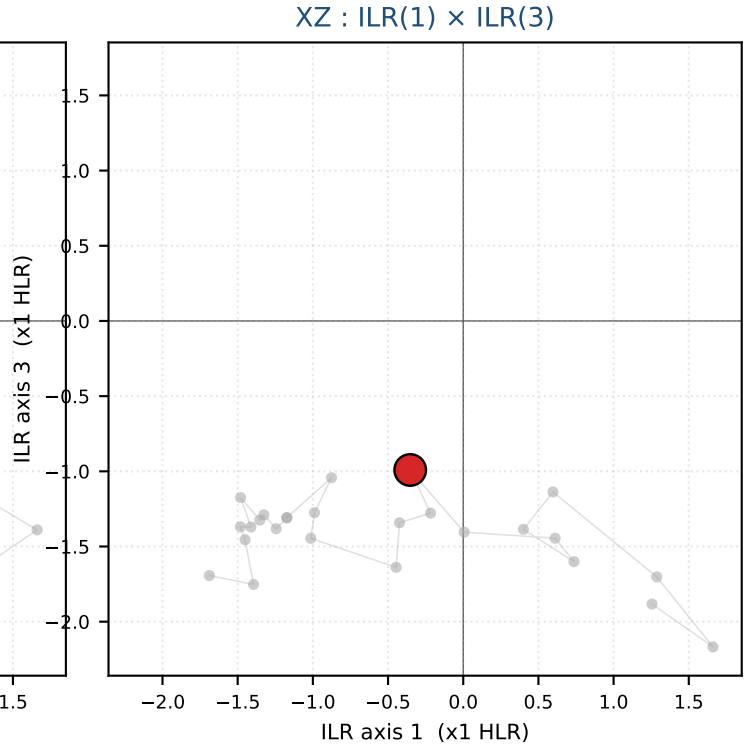
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=17 | 2017 | D=9 T=26 | level 1 first-principles

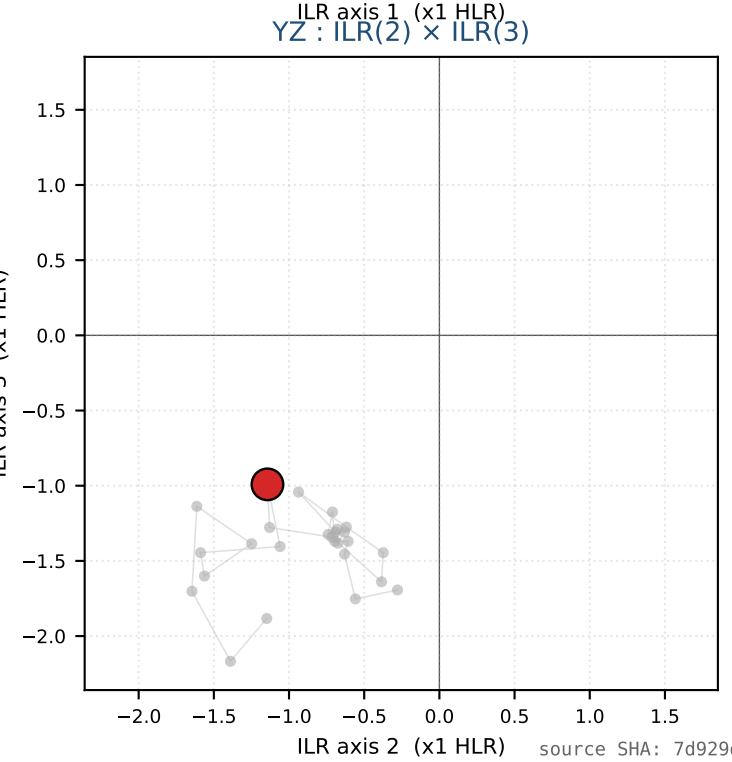
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



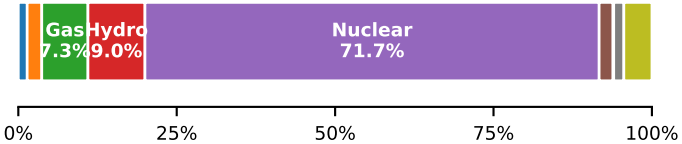


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0140	-0.799
Coal	0.0231	-0.300
Gas	0.0729	+0.851
Hydro	0.0900	+1.061
Nuclear	0.7168	+3.137
Other Fossil	0.0219	-0.352
Other Renewabl	0.0012	-3.281
Solar	0.0160	-0.668
Wind	0.0443	+0.353



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

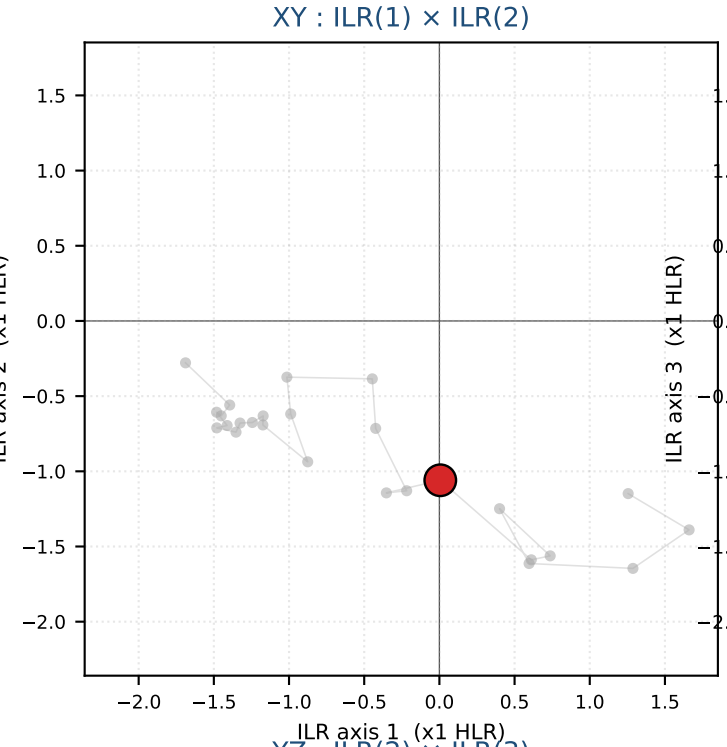
display factor: x1 (raw HLR x 1)

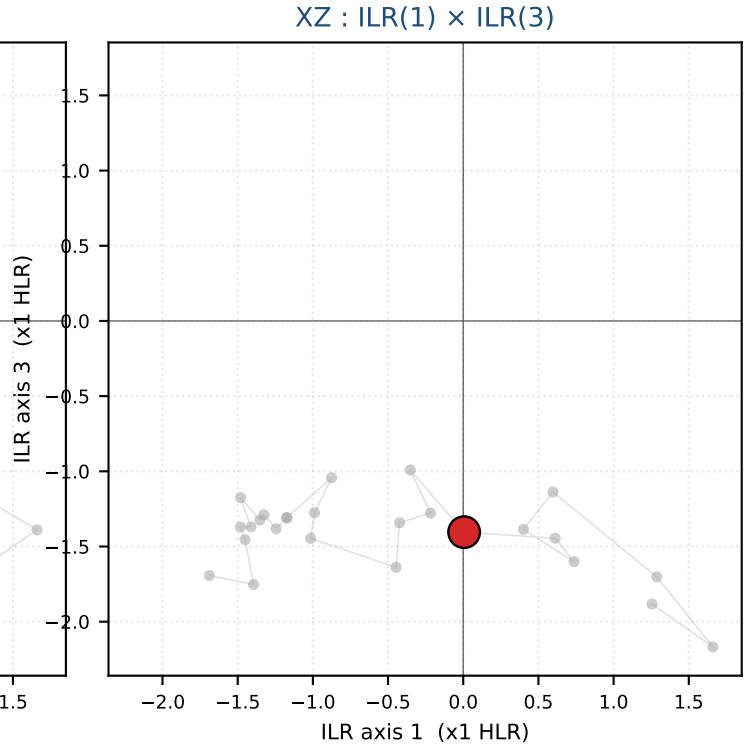
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=18 | 2018 | D=9 T=26 | level 1 first-principles

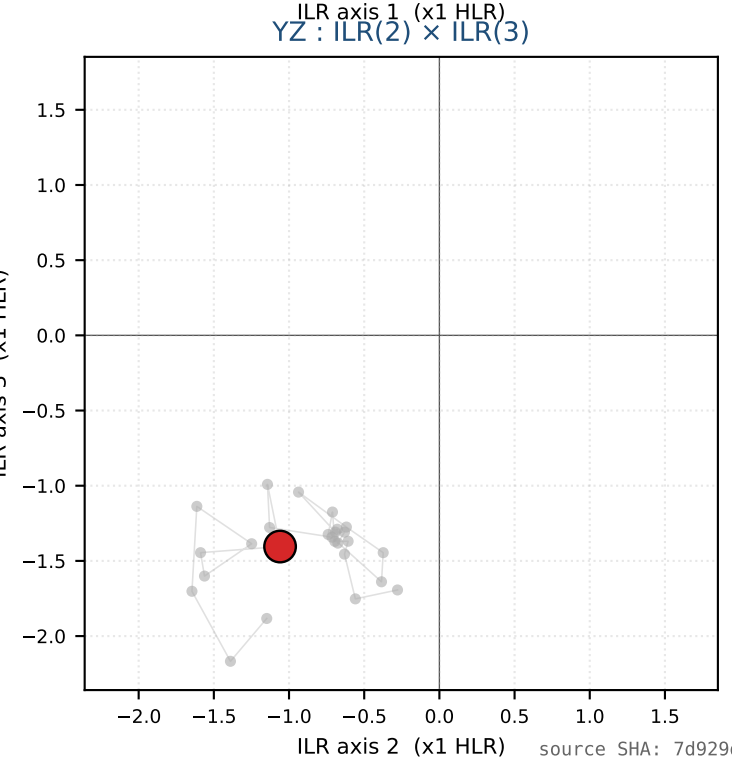
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



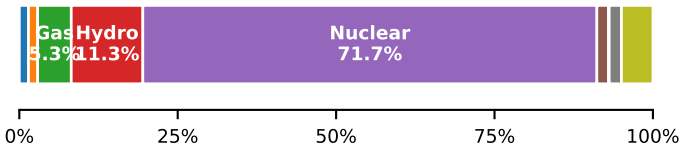


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0145	-0.707
Coal	0.0144	-0.715
Gas	0.0530	+0.587
Hydro	0.1130	+1.344
Nuclear	0.7166	+3.191
Other Fossil	0.0184	-0.473
Other Renewabl	0.0011	-3.327
Solar	0.0193	-0.421
Wind	0.0496	+0.521



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

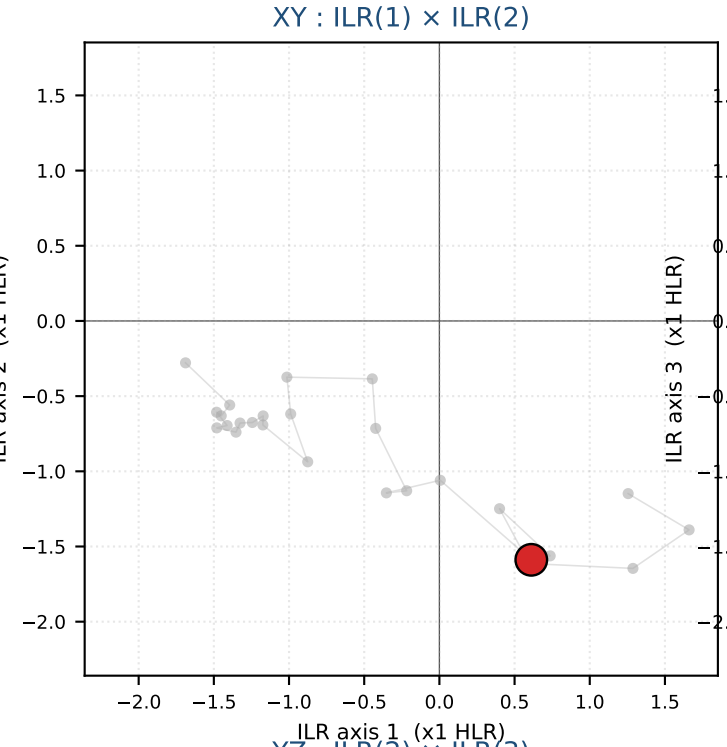
display factor: x1 (raw HLR x 1)

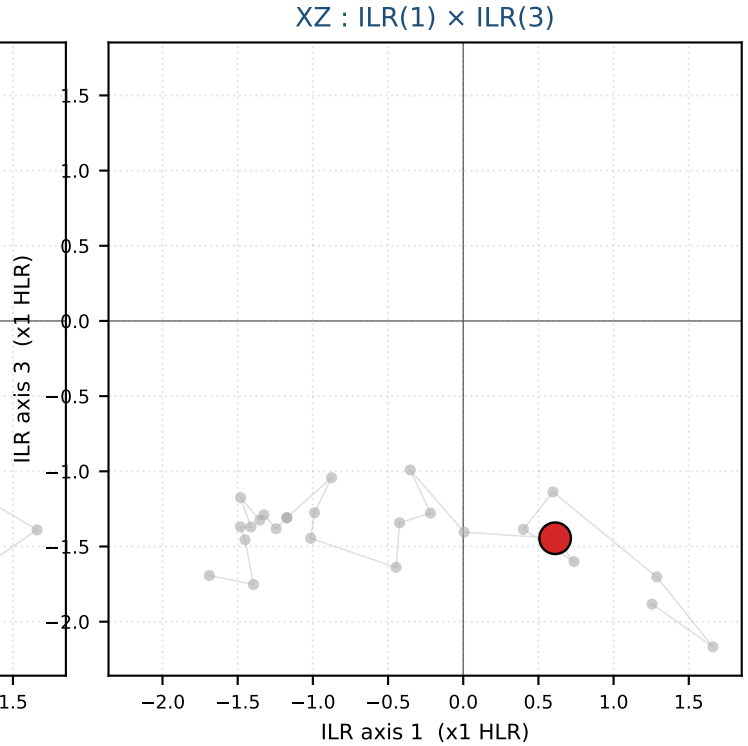
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=19 | 2019 | D=9 T=26 | level 1 first-principles

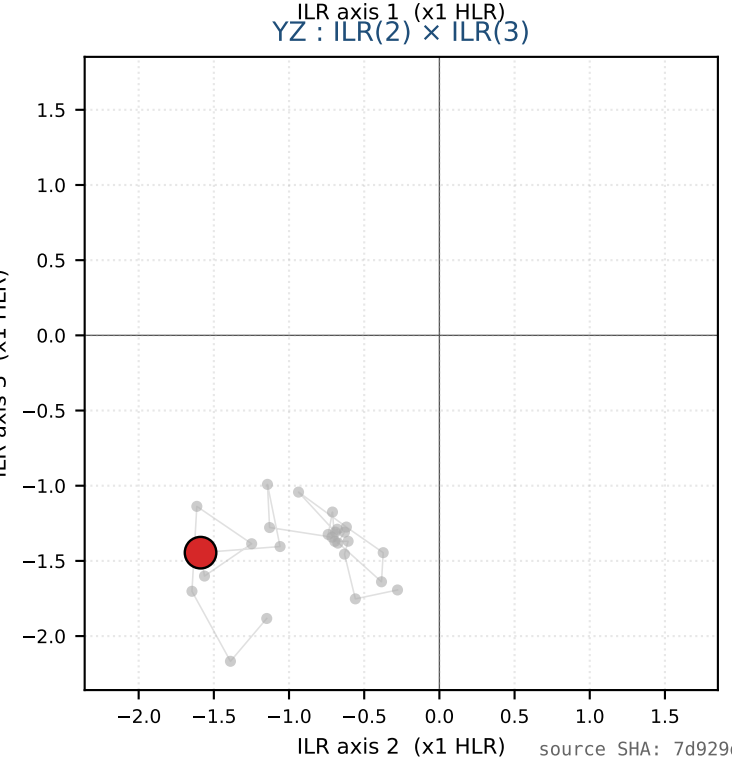
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



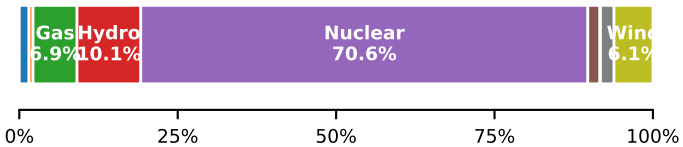


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0153	-0.628
Coal	0.0064	-1.493
Gas	0.0694	+0.885
Hydro	0.1006	+1.257
Nuclear	0.7056	+3.204
Other Fossil	0.0190	-0.412
Other Renewabl	0.0011	-3.279
Solar	0.0213	-0.298
Wind	0.0614	+0.763



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

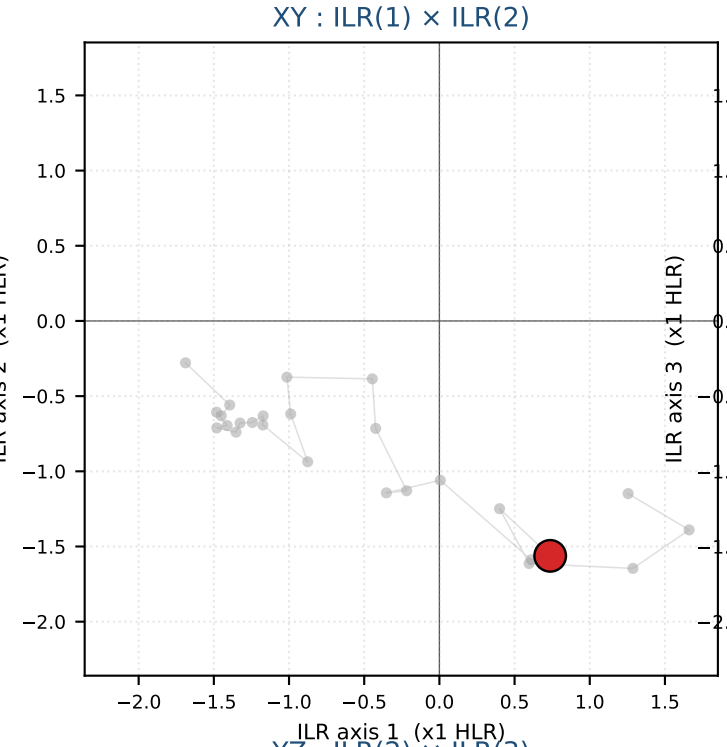
display factor: x1 (raw HLR x 1)

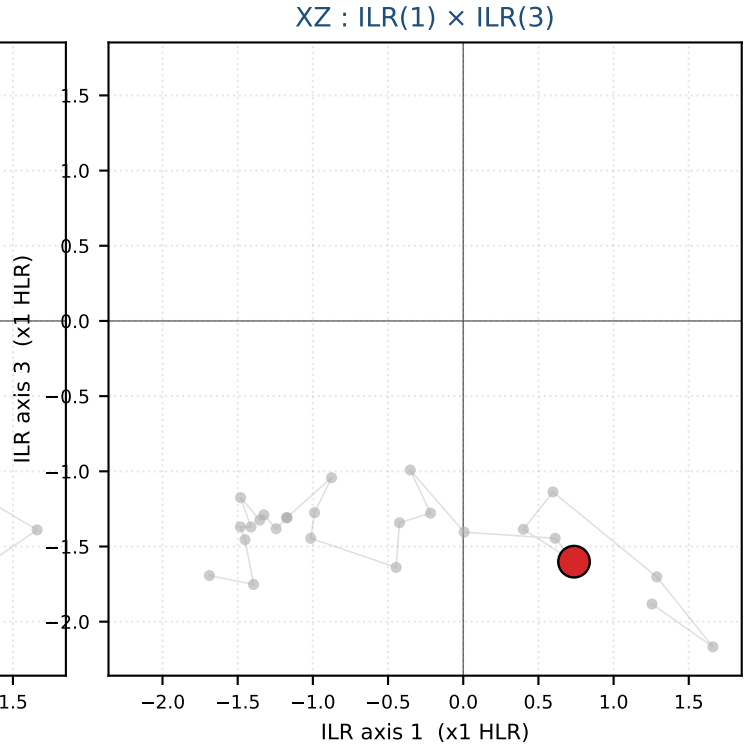
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=20 | 2020 | D=9 T=26 | level 1 first-principles

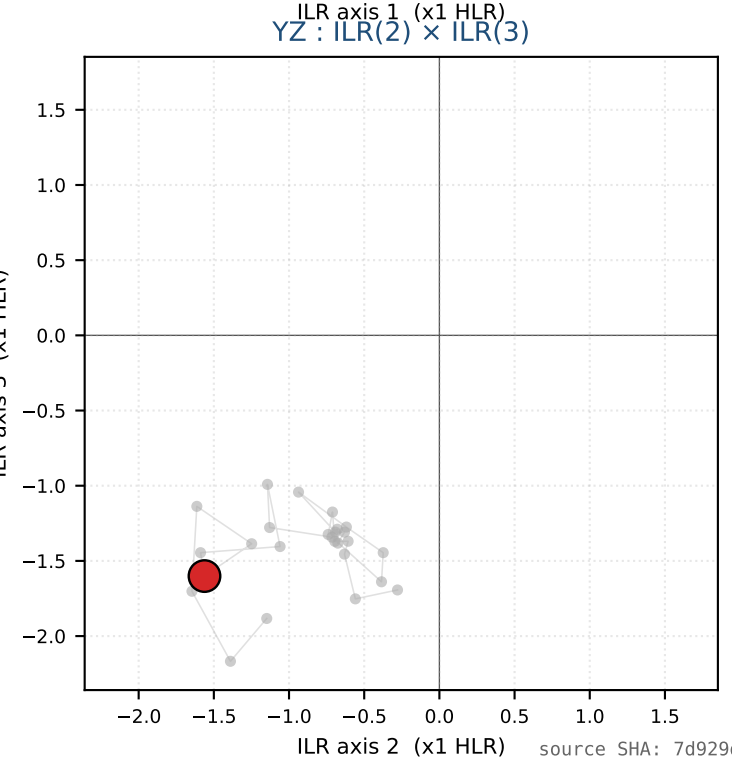
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



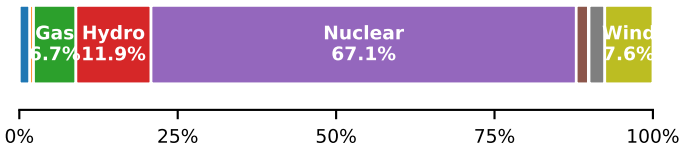


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0166	-0.601
Coal	0.0059	-1.643
Gas	0.0669	+0.791
Hydro	0.1186	+1.364
Nuclear	0.6711	+3.098
Other Fossil	0.0191	-0.464
Other Renewabl	0.0012	-3.265
Solar	0.0248	-0.199
Wind	0.0760	+0.919



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

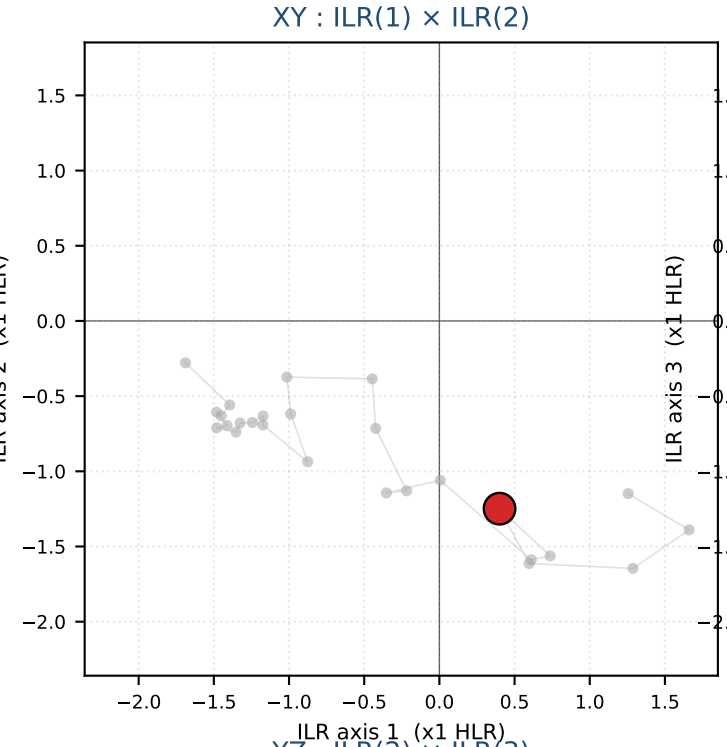
display factor: x1 (raw HLR x 1)

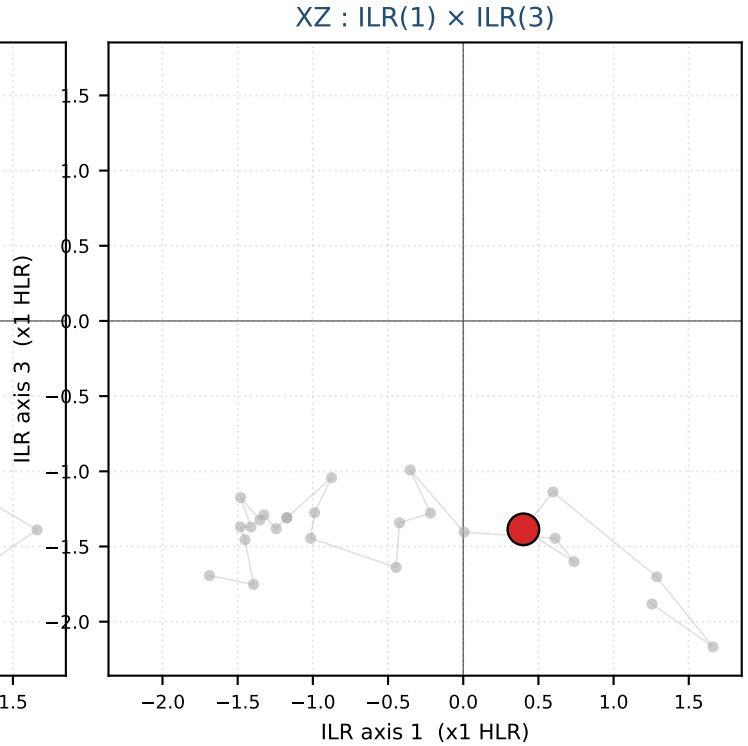
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=21 | 2021 | D=9 T=26 | level 1 first-principles

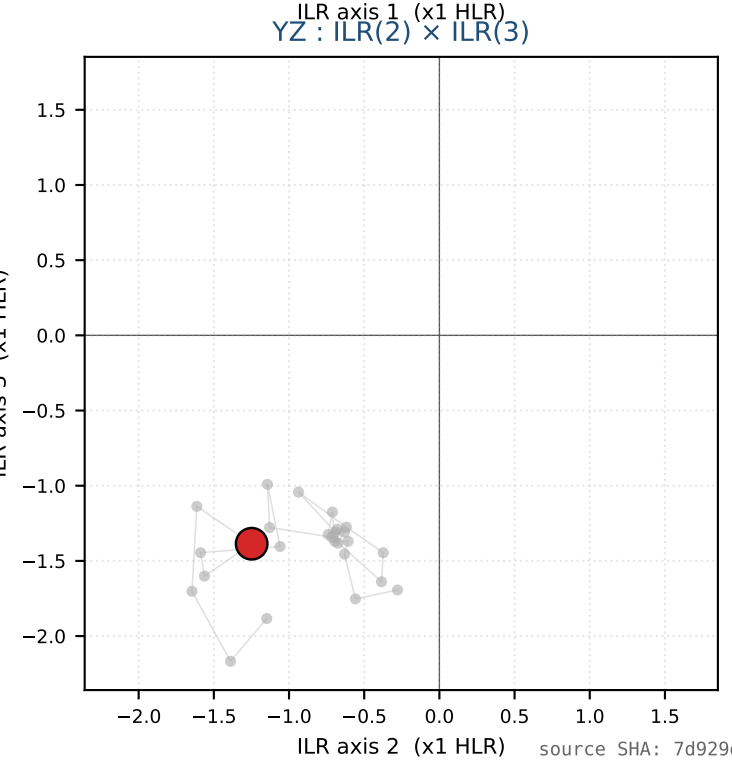
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



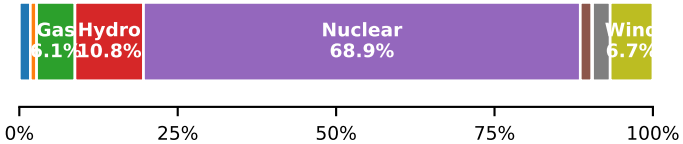


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0174	-0.584
Coal	0.0099	-1.150
Gas	0.0605	+0.662
Hydro	0.1082	+1.243
Nuclear	0.6894	+3.095
Other Fossil	0.0184	-0.531
Other Renewabl	0.0011	-3.388
Solar	0.0278	-0.117
Wind	0.0675	+0.770



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

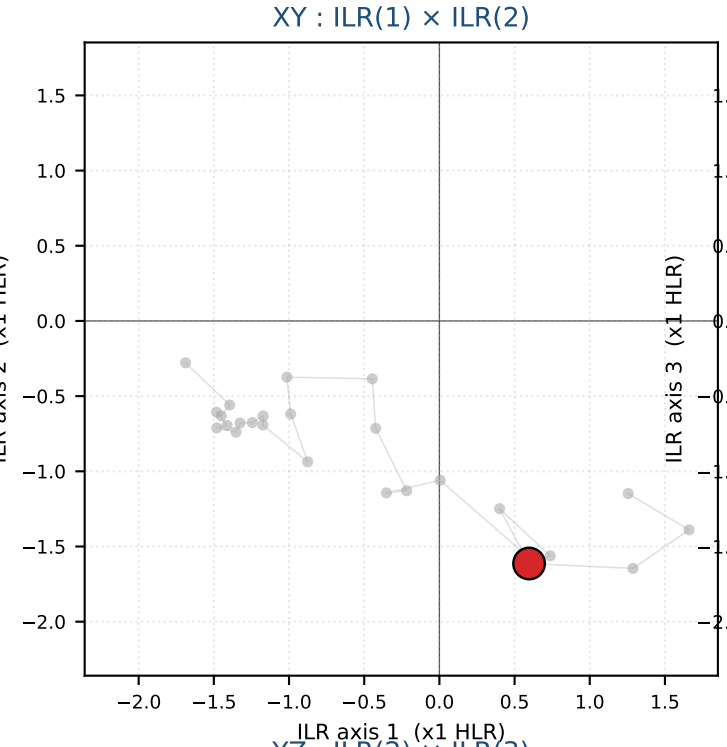
display factor: x1 (raw HLR x 1)

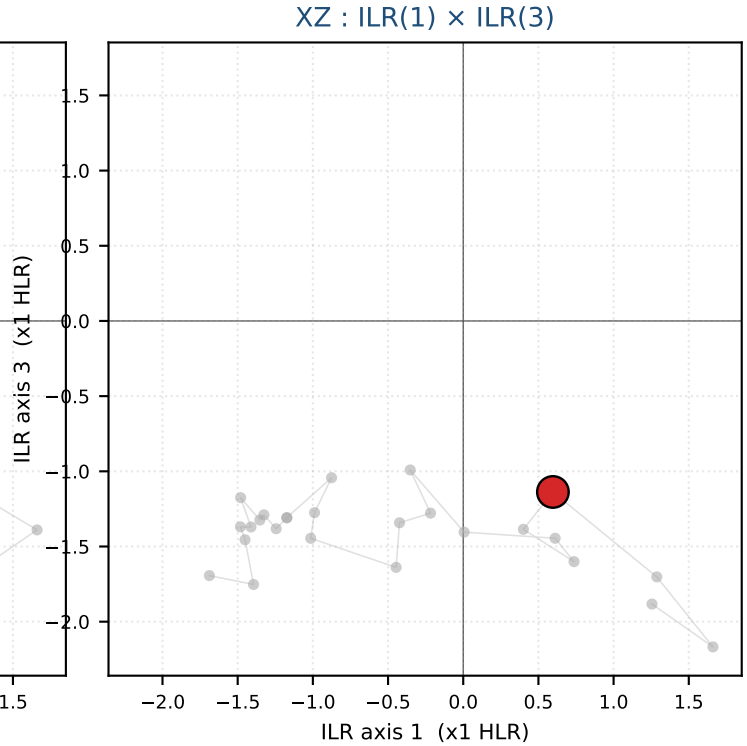
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=22 | 2022 | D=9 T=26 | level 1 first-principles

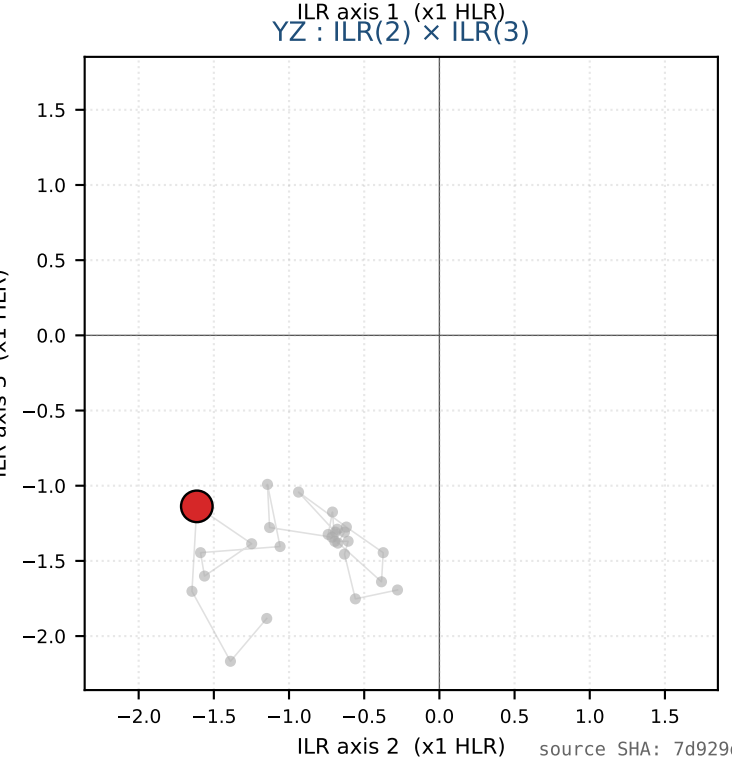
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



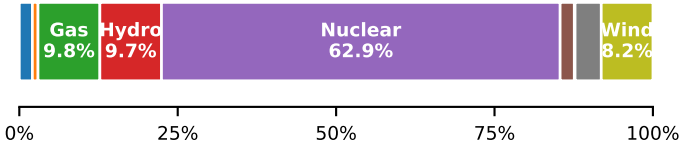


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0206	-0.560
Coal	0.0089	-1.404
Gas	0.0976	+0.994
Hydro	0.0973	+0.990
Nuclear	0.6292	+2.857
Other Fossil	0.0223	-0.483
Other Renewabl	0.0013	-3.340
Solar	0.0413	+0.133
Wind	0.0815	+0.814



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

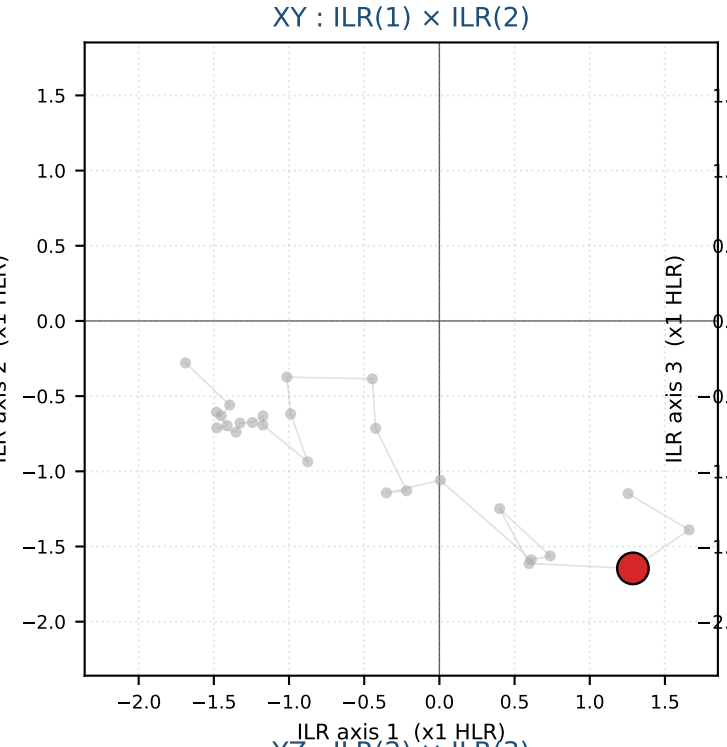
display factor: x1 (raw HLR x 1)

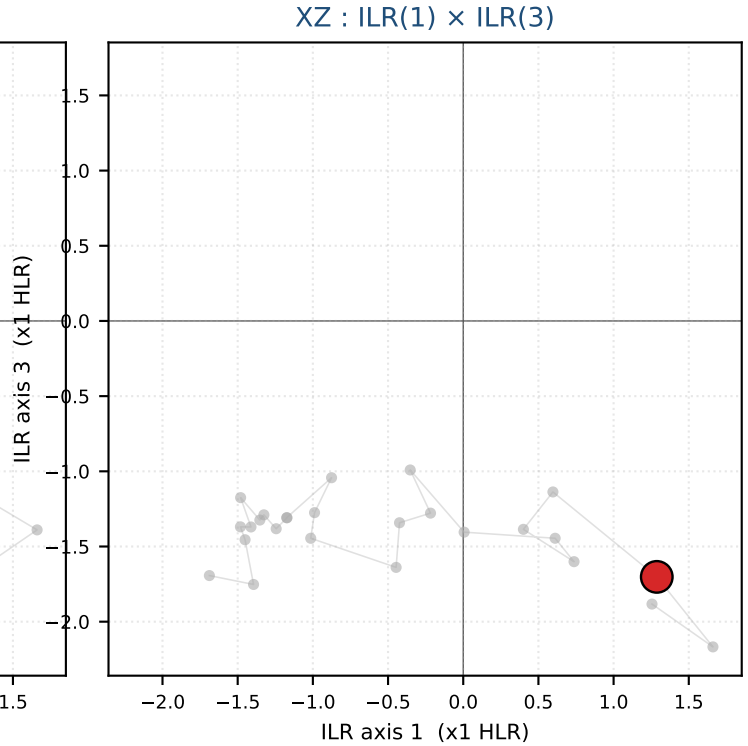
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=23 | 2023 | D=9 T=26 | level 1 first-principles

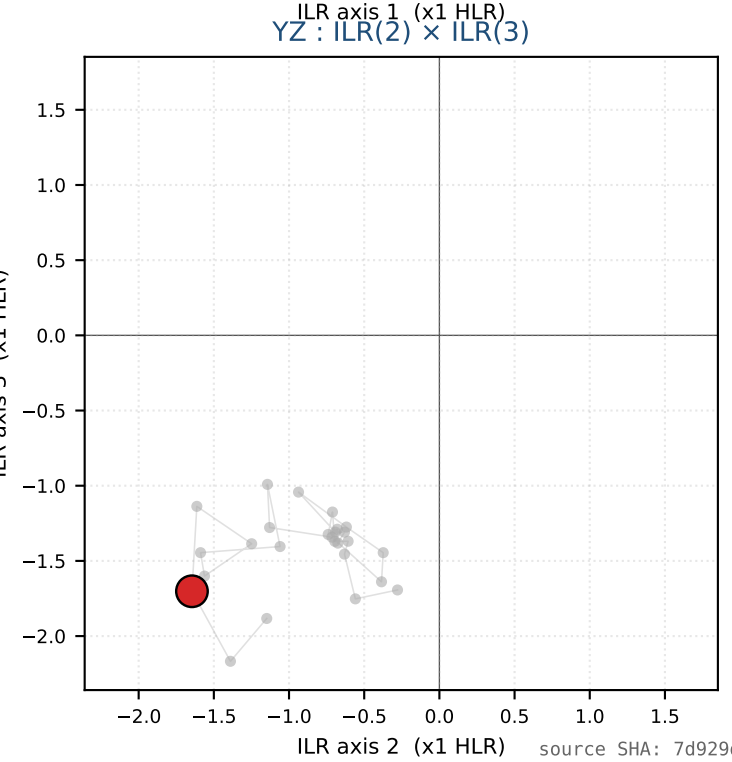
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



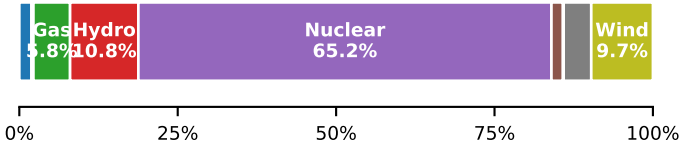


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0192	-0.455
Coal	0.0031	-2.275
Gas	0.0579	+0.651
Hydro	0.1078	+1.272
Nuclear	0.6520	+3.073
Other Fossil	0.0182	-0.507
Other Renewabl	0.0011	-3.296
Solar	0.0435	+0.366
Wind	0.0973	+1.170



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

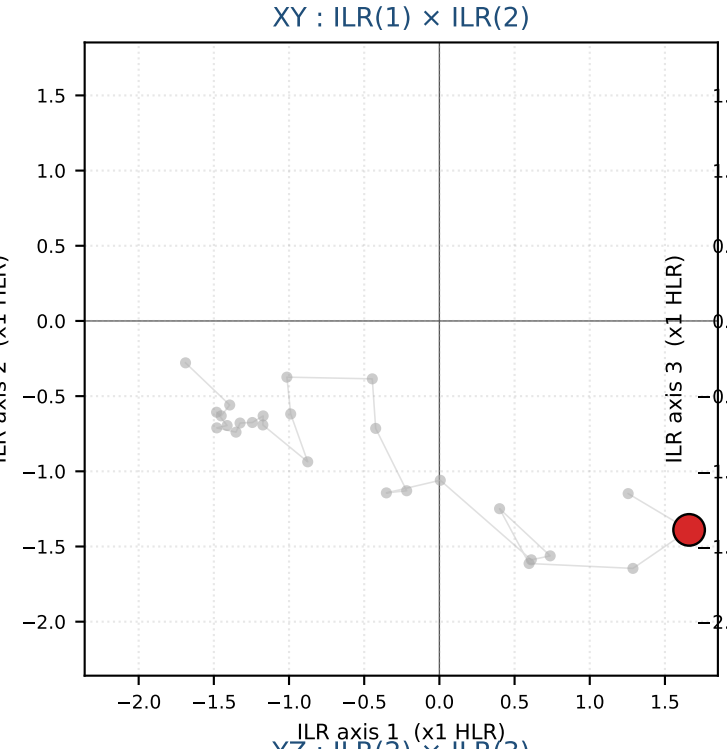
display factor: x1 (raw HLR x 1)

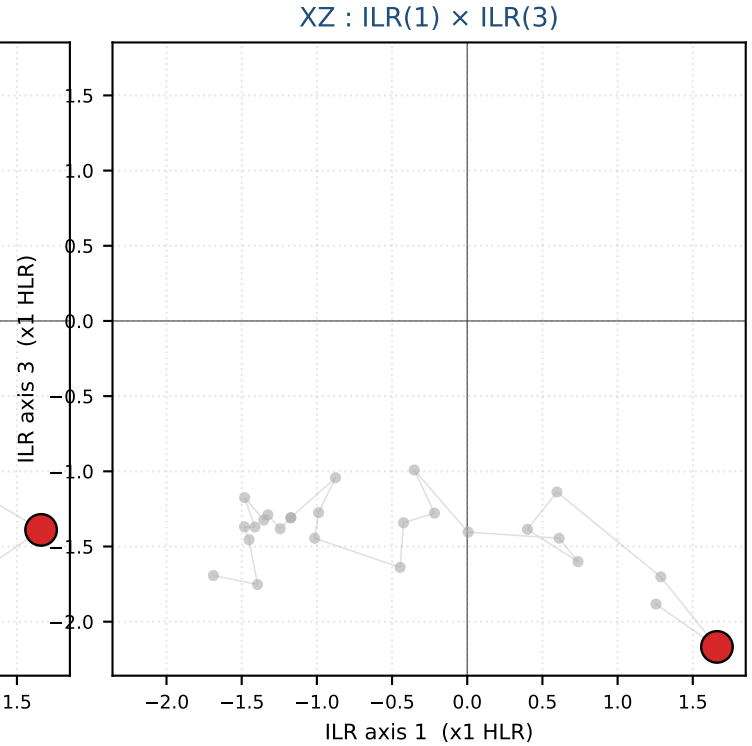
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=24 | 2024 | D=9 T=26 | level 1 first-principles

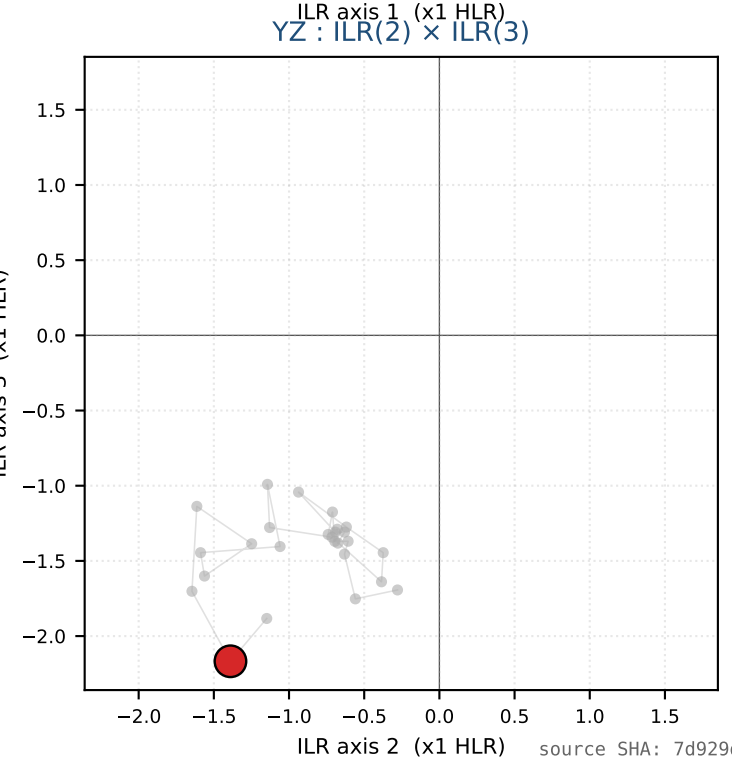
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

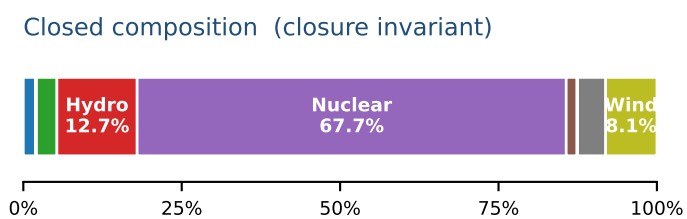




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0190	-0.322
Coal	0.0018	-2.671
Gas	0.0322	+0.206
Hydro	0.1266	+1.574
Nuclear	0.6772	+3.251
Other Fossil	0.0170	-0.435
Other Renewabl	0.0010	-3.252
Solar	0.0443	+0.523
Wind	0.0809	+1.126





closure: Σ = 1.000000 OK

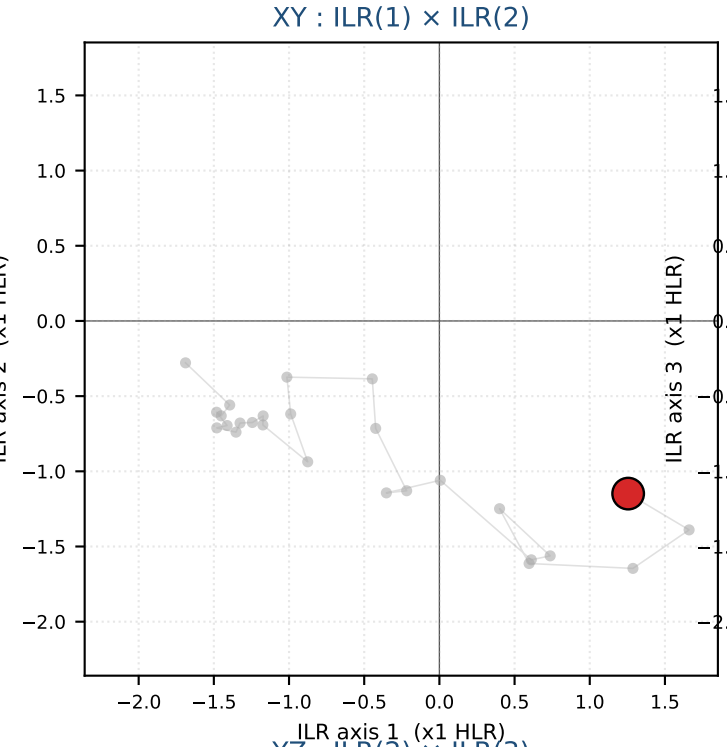
display factor: x1 (raw HLR x 1)

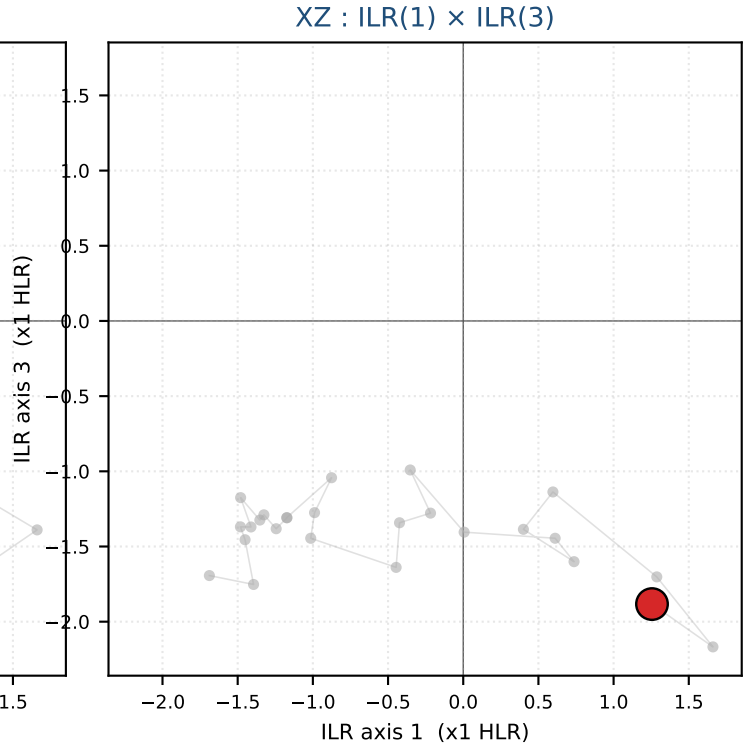
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=25 | 2025 | D=9 T=26 | level 1 first-principles

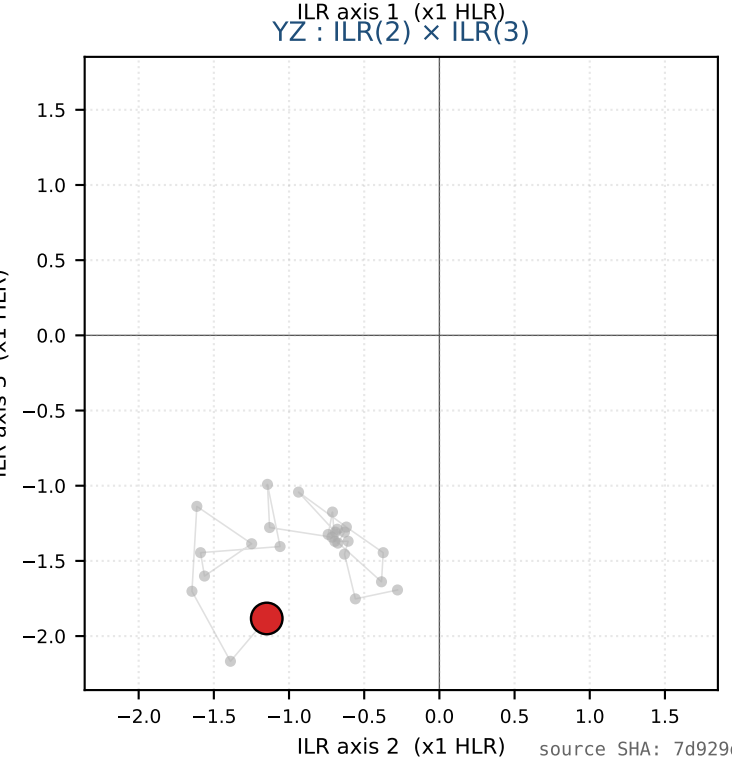
ember_fra | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

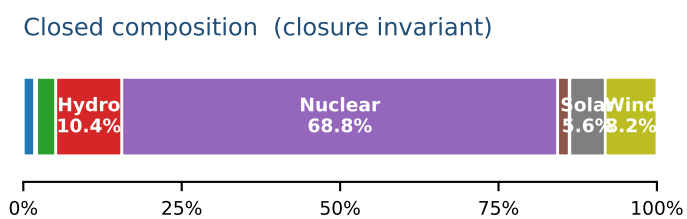




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0180	-0.433
Coal	0.0031	-2.209
Gas	0.0303	+0.085
Hydro	0.1042	+1.322
Nuclear	0.6879	+3.209
Other Fossil	0.0182	-0.424
Other Renewabl	0.0010	-3.325
Solar	0.0558	+0.698
Wind	0.0816	+1.077





closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

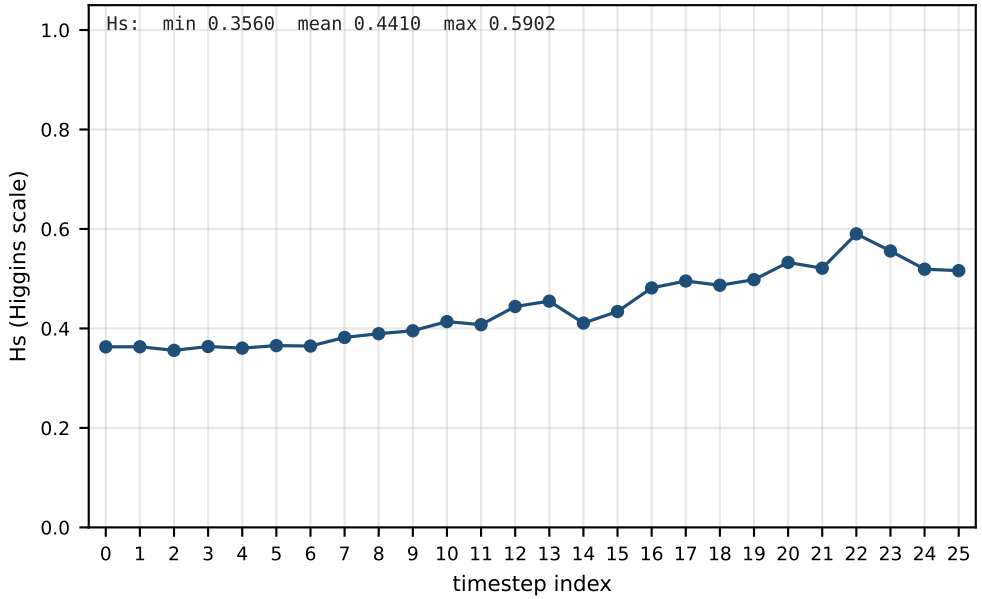
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

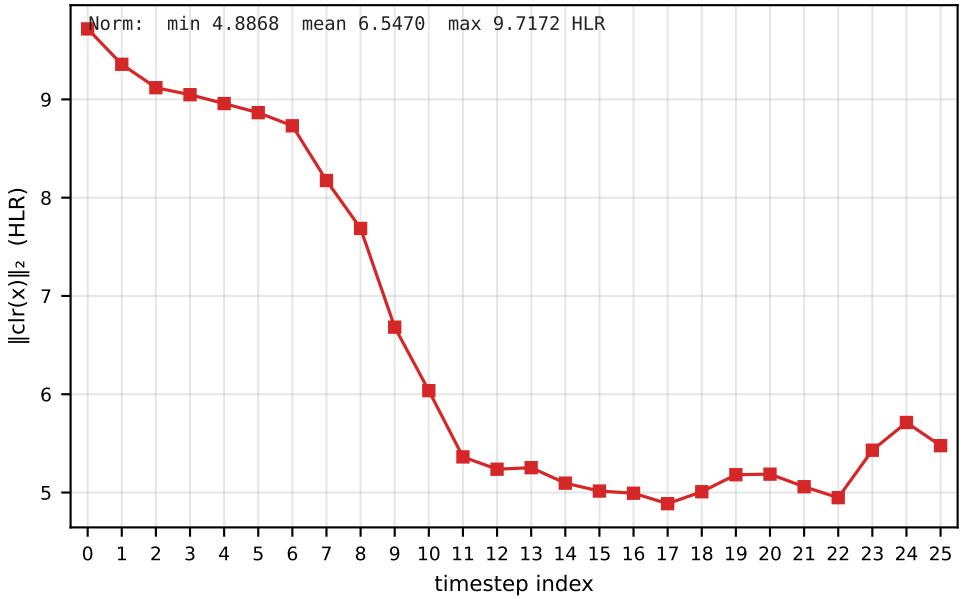
STAGE 1 SUMMARY | ember_fra | D=9 T=26 | Order 1 (first-principles closure)

Engine cnt 2.0.3 | Schema 2.0.0 | Generated 2026-05-05T20:04:42

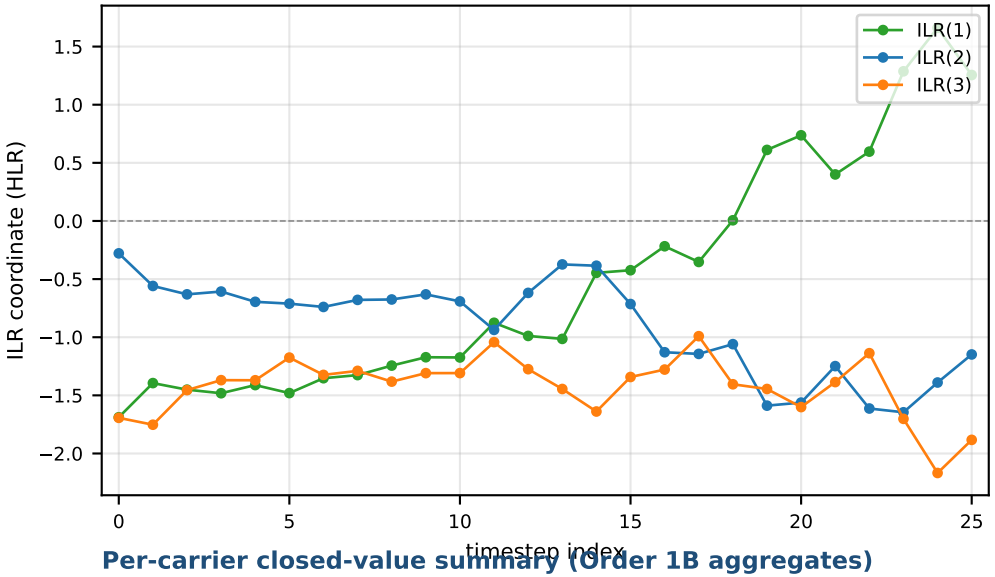
Hs trajectory — single-composition entropy / ln(D)



Aitchison norm — distance from centroid (HLR)



ILR axes 1–3 trajectories (Helmert basis)



Order-1 invariants (whole trajectory)

```
closure invariant     :  max |Σ closed − 1| = 2.22e-16
closure status        :  OK ✓

Hs range              :  [0.3560, 0.5902]
Aitchison norm range  :  [4.8868, 9.7172] HLR
ILR(1) range          :  [-1.6883, +1.6607]
ILR(2) range          :  [-1.6456, -0.2786]
ILR(3) range          :  [-2.1677, -0.9910]

Source file           :  ember_FRA_France_generation_TWh.csv
Source SHA-256        :  7d929d9c4c7d1b47754582e586f88b44...
Closed-data SHA-256   :  f46d758c991ee166ef1b08f4edc5b02e...
Content SHA-256       :  e7875f1a368d0d3729f498da7a8691c7...
Ordering              :  temporal=True, method=by-time
Zero replacement δ    :  1e-15
```

Per-carrier closed-value summary (Order 1B aggregates)

#	carrier	min	mean	max	std
0	Bioenergy	0.0047	0.0110	0.0206	0.0050
1	Coal	0.0018	0.0278	0.0507	0.0164
2	Gas	0.0216	0.0454	0.0976	0.0174
3	Hydro	0.0806	0.1066	0.1338	0.0118
4	Nuclear	0.6292	0.7413	0.7936	0.0483
5	Other Fossil	0.0167	0.0195	0.0224	0.0016
6	Other Renewabl	0.0008	0.0010	0.0013	0.0001
7	Solar	0.0000	0.0137	0.0558	0.0164
8	Wind	0.0001	0.0339	0.0973	0.0307